RATIONAL REFLECTIONS AND PALINDROMIC WORDS IN THE COHN MONOID

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Abstract. We study the rational reflection that a palindrome p defines for the two Cohn matrices. We classify the finite words whose terminal columns it takes to other word columns: they are exactly the concatenations of the blocks $ab\,p$ and $ba\,p$, which it exchanges. Hence two palindromes have equal Gram labels and a root midpoint at the ratio of the centre exactly when they are exchanged around it. Such pairs have nonprimitive completed letter counts, so no two central words collide at such a midpoint; an explicit quadruple shows that this hypothesis cannot be dropped. Exchanges around nested centres compose into cubes of palindromes with one label. In the Thue–Morse coding of Reutenauer and Vuillon, the Gram label of a palindrome is the length of an explicit Christoffel word and the Gram root its number of letters b; for Markov words, uniqueness holds if and only if equal labels force directing words of equal length. An exact enumeration of all palindromes with label below 10^{26} finds no coincidence of labels that exchanges do not produce. Beyond it, two palindromes of 52 letters share a label without being joined by exchanges, yet both are built around the empty word; we conjecture that any two palindromes with the same label are built around a common palindrome, which would imply Frobenius–Markov uniqueness. Away from the midpoint we derive the two-sided reflection equation and a fourth-power bound on the common-ancestor label under bounded parent imbalance. The uniqueness conjecture itself remains open.

1. Introduction

The Markov equation

$$x^2 + y^2 + z^2 = 3xyz$$

has a tree of positive integer solutions, and the Frobenius–Markov uniqueness conjecture asks whether a solution is determined, up to permutation, by its largest entry. One classical way to encode a solution is by a word: Christoffel words in two letters parametrize the tree, and under the Cohn representation the largest entry of a triple becomes a matrix invariant of the corresponding central word. A hypothetical pair of distinct triples with the same largest entry would then be a pair of distinct central words with the same invariant, and one can look for structures in the word setting that forbid such a coincidence. This article isolates one such structure and determines it completely.

The structure is a reflection. For the two Cohn matrices

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix},$$

a positive matrix word carries three kinds of information at once: a literal binary word, an integral Gram matrix, and the arithmetic of a root of -1 modulo its first diagonal entry. Each description makes some operations transparent and hides others. A reflection acts simply on Gram roots, but its compatibility with positive words turns out to be far more restrictive, and the question we answer is exactly how restrictive.

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The distinction is already visible in a small example. The palindromes *baab* and *abba* have Gram matrices

$$\begin{pmatrix} 1130 & 467 \\ 467 & 193 \end{pmatrix}, \qquad \begin{pmatrix} 1130 & 437 \\ 437 & 169 \end{pmatrix}.$$

They have the same label and their normalized roots have midpoint $2/5$. There is no numerical approximation here. The coincidence itself has been noticed before, as an equality of matrix entries [9, p. 4] and as a counterexample to a strengthening of the uniqueness conjecture for the values of lattice paths [8, p. 28]. Yet their completed letter counts are $(3, 3)$, which rules out central words and hence the usual Christoffel interpretation. In the Thue–Morse coding of Reutenauer and Vuillon, recalled in section 2.4, the two palindromes become *abaaaba* and *aababaa*, and the equality of their labels becomes the equality of the lengths of two central words; it is the first such coincidence. The issue is to determine whether this failure of primitivity is incidental to the example or forced by the reflection.

We prove that it is forced, and classify every pair responsible for it. A palindrome p determines a rational reflection R_p, which fixes the terminal column $e = (2, 1)^{\mathsf{T}}$ and is orthogonal for the Gram form of p. The principal word theorem states that, for all finite words x and y,

$$[R_p\mu(x)e] = [\mu(y)e] \quad \Longleftrightarrow \quad x \in \{ab\,p,\ ba\,p\}^*, \quad y = \widehat{x},$$

where brackets denote a projective line and the hat exchanges the blocks $ab\,p$ and $ba\,p$ (theorem 4.11). This is a classification for arbitrary finite words, uniformly in the centre, and the projective equality always lifts to exact equality of columns. When p is the central word g_d of a standard directive d, the blocks are the images of ab and ba under the standard morphism ϕ_d, and the theorem reads $[R_d\nu_d(x)e] = [\nu_d(y)e]$ if and only if $x \in \{ab, ba\}^*$ and $y = \widehat{x}$, with $\nu_d = \mu \circ \phi_d$ (theorem 3.1). For this case we give a second proof by projective cylinders, which also measures how far a reflected column can move.

For palindromes, the word theorem has a stronger arithmetic consequence. Equal labels M and the midpoint condition

$$\frac{\rho_X + \rho_Y}{2M} = \frac{\rho_p}{M_p}$$

are equivalent to a full matrix conjugacy by R_p, and the decoder then forces the two palindromes to be $p\,X$ and $p\,\widehat{X}$, with X a concatenation of the blocks (corollary 4.12): no common prefix needs to be assumed, and the prefix p is forced. If X has $2k$ blocks, the completed counts of both palindromes are $2k + 1$ times those of p, so they are not primitive when $k > 0$. For a standard ancestor this gives the preimages $\widetilde{w}w$ and $\widetilde{\widehat{w}}\,\widehat{w}$, with w a concatenation of ab and ba, and completed counts $(2k + 1)L_d(1, 1)^{\mathsf{T}}$ (theorem 4.2). Thus the reflection creates an infinite family of exact palindromic label equalities while excluding every nontrivial central pair at the midpoint of any palindrome.

We also retain the information that changes when the midpoint moves. The conjugacy becomes a two-sided matrix equation with a rank-one term, and its action on a centered reciprocal cusp coordinate is an affine reflection. For a hypothetical pair of distinct central words with equal labels, reduced integral coordinates for the midpoint supply a nonzero integer defect, and the first-letter cylinders of the standard generators convert that defect into a quantitative restriction on the common ancestor. The precise statement is theorem 5.4: when the two parent traces are comparable, the ancestor label M_0 is bounded by a constant multiple of $M^{1/4}$, the constant being explicit in the data of the actual pair, with no factorization of M required and no restriction on the parity of M. Along a one-sided directive the two parents grow apart, the constant degrades, and the bound weakens to a cubic scale. Neither regime excludes all collisions.

The block exchange around a palindrome p (proposition 4.5) has a simple arithmetic: the label of p divides the common label, and the exchange conjugates the complementary Gaussian factor (proposition 4.6). Exchanges around nested centres compose. When the centre of an exchange is itself one of two exchanged palindromes, the four words built around them form a square of palindromes with one label, and r nested levels form a cube with 2^r vertices (theorem 4.14). The pairs of such a square that are not exchanges have their root midpoints shifted off every centre, by exactly half the width of a short rung (corollary 4.15). In the Thue–Morse coding the label is the length of an explicit Christoffel word and the root its number of letters b (proposition 2.6). An exact enumeration of all palindromes with label below 10^{26} finds no coincidence of labels that the exchanges do not produce, and every class of equal labels it finds is a single exchange or a square. Beyond that range two palindromes of 52 letters share a label without being joined by exchanges (remark 6.2), but both are structured around the empty word, and conjecture 6.1, whose proof would imply the uniqueness conjecture, asks that any two twins be structured around a common palindrome. Twins on a reflection axis, and all twins of the enumeration, have directing words of the same length in the Thue–Morse coding. This length is half the sum of the partial quotients of the label divided by the root, and for Markov words the fixed-sum theorem of Lagisquet, Pelantová, Tavenas and Vuillon [8] turns the equality of lengths into an equivalent form of the uniqueness conjecture (proposition 6.5).

The prescribed midpoint is essential: a general equal-label pair need not have it (remark 4.7 exhibits one), and away from it the inequalities above remain necessary conditions rather than exclusions. What the article contributes is the complete word classification at a specified reflection, together with a quantitative account of the information that survives off its axis. The Frobenius–Markov uniqueness conjecture itself is not resolved here.

1.1. **Relation to earlier work.** The standard morphisms and palindromic extensions used here belong to classical Sturmian combinatorics. Justin's functional identity relates iterated palindromic closure to composition of standard morphisms [6, Lemma 2.1]. Its left and right forms, with the same conventions used below, appear in Perrin and Reutenauer [11, Section 2]. The incidence formulas of Berthé, de Luca and Reutenauer describe the completed letter counts of central words and their standard factorizations [2, Section 3]. These results supply the word transport and the primitive-count test in our application. The involution obtained by reversing a directive in [2] changes the central word by Christoffel duality; our reflection instead acts on a positive matrix orbit while the directive remains fixed.

The language that appears in the answer is also familiar. If $\theta(a) = ab$ and $\theta(b) = ba$, then $(ab \mid ba)^* = \theta(\{a, b\}^*)$ is the image of the Thue–Morse substitution. Reutenauer and Vuillon use this substitution together with palindromic and antipalindromic closure to construct Markov numbers [13]: in their coding a Markov number is two more than the length of $\mathrm{Pal}(\theta(\mathrm{Pal}(av)))$, and the injectivity of this length is equivalent to the uniqueness conjecture [13, Theorem 1]. Proposition 2.6 identifies that length plus two, the length of the Christoffel word $a\,\mathrm{Pal}(\theta(F_a(z)))b$, with our Gram label, for every palindrome z. Abram, Lapointe and Reutenauer extend the construction to whole Markov triples [1]. Our conclusion concerns the role of this language in a different problem: it exhausts the finite words whose positive terminal vectors are carried to another such vector by the prescribed rational reflection. The proof establishes this exhaustion uniformly for the reflection of every palindrome, before imposing palindromicity on the reflected words.

Palindromic symmetry is well established in the geometry of two-generator groups. Kassel and Reutenauer prove that a cyclically reduced basis with both words of odd

length has a unique palindromic conjugate basis [7]. Gilman and Keen relate palindromic words to axes perpendicular to the common perpendicular of the generator axes [4]. These results concern primitive bases or geometric axes. The decoder here retains the positive word, the finite terminal vector and the fixed rational reflection, and applies to every finite word, including nonprimitive ones.

The distinction between a full matrix and one of its entries matters when connecting the result to Markov arithmetic. Reutenauer's Christoffel correspondence is a bijection with whole proper Markov triples [12, Theorem 1]. More recently, Veselov identifies Markov fractions with Cohn matrix indices and describes continued fractions through a mirrored Conway topograph [14, Sections 3–4]. Jouteur, Paris-Romaskevich and Thomas characterize modular involutions exchanging rational pairs and study Springborn operations, including a q-deformed midpoint formula for Farey neighbors [5, Sections 5–7]. Those constructions retain a tree position, an arithmetic regularity condition or a whole rational fraction. They do not identify which arbitrary positive words survive reflection in the orbit considered here.

There are also injectivity theorems for restricted word languages. Lapointe and Reutenauer prove that a Cohn matrix entry is strictly increasing in military order on the language of one fixed Sturmian sequence [9, Corollary 1]. Our comparison allows arbitrary palindromes, whose completions need not lie in one such language. Their Gram labels are positive integers, and are Markov numbers under the additional Christoffel hypothesis. Equal labels and a specified root midpoint recover a full Gram congruence, which brings the arbitrary-word decoder into play. This yields an exact classification of the reflected pairs and shows why every nonempty pair fails the primitive completed-count condition. No claim is made that an arbitrary equal-label pair has the prescribed midpoint.

1.2. **Organization.** Section 2 fixes the literal standard morphisms, completed counts and Gram normalization, and reads the label and the root in the Thue–Morse coding. Section 3 proves the finite-word decoder by separating projective cylinders and reading two letters at a time. Section 4 derives the palindromic classification and the central-word obstruction, extends the construction, the decoder and the classification to every palindromic centre, shows that the midpoint hypothesis cannot be removed, and proves that nested exchanges produce cubes of twins. Section 5 treats moving midpoints, explains the weighted scale that must be retained, and proves the common-ancestor bound. Section 6 states the conjecture on twins, explains the ladders in which they appear, reads the lengths of the directing words, and isolates the remaining mathematical question.

2. Standard words and their Gram matrices

We first fix the word conventions. This is essential because two standard morphisms can have the same incidence matrix and different literal images. Throughout, a word is a finite word over $\{a, b\}$, $\widetilde{z}$ denotes reversal, and ε denotes the empty word. A palindrome satisfies $z = \widetilde{z}$. Products of matrices follow the order of letters from left to right.

2.1. **Palindromic closure and standard morphisms.** For a word z, let z^+ be the shortest palindrome having z as a prefix. The iterated right palindromization map is defined by

$$\mathrm{Pal}(\varepsilon) = \varepsilon, \qquad \mathrm{Pal}(dc) = (\mathrm{Pal}(d)c)^+, \quad c \in \{a, b\}.$$

For example, $\mathrm{Pal}(ab) = aba$ and $\mathrm{Pal}(aba) = abaaba$. We call the values of this map *central words*. This is the usual binary central-word class; its relation with Christoffel words is recalled below [2, Section 3]. The word d is called a *directive*.

Use the right standard morphisms

$$\phi_a:\ a\mapsto a,\ b\mapsto ba, \qquad \phi_b:\ a\mapsto ab,\ b\mapsto b, \qquad \phi_{dc}=\phi_d\circ\phi_c. \tag{1}$$

The corresponding left morphisms are $\psi_a(a)=a$, $\psi_a(b)=ab$, $\psi_b(a)=ba$, and $\psi_b(b)=b$. Define, for every word z,

$$F_c(z)=c\phi_c(z)=\psi_c(z)c, \qquad F_{dc}=F_d\circ F_c, \qquad F_\varepsilon(z)=z.$$

The equality in this definition follows by moving the final c across the image of each letter. Reversing it shows that F_c preserves palindromes. Justin's functional identity [6, Lemma 2.1], in its right-morphism form [11, Section 2], identifies $F_d(\varepsilon)$ with $\mathrm{Pal}(d)$; in the present convention it gives

$$F_d(z)=g_d\phi_d(z), \qquad g_d=\mathrm{Pal}(d), \qquad F_d(\mathrm{Pal}(t))=\mathrm{Pal}(dt). \tag{2}$$

The first equality also follows directly by induction on $|d|$ once $F_d(\varepsilon)=g_d$ is known. Thus F_d extends the action on central words to every binary palindrome.

The completed-count formulas below are the incidence form of the standard central-word construction [2, Corollaries 3.1–3.2]. We include the short proof to fix the composition convention.

Lemma 2.1 (Literal identities and completed counts)**.** *For every directive d,*

$$\phi_d(ab)=abg_d, \qquad \phi_d(ba)=bag_d. \tag{3}$$

Let L_d be the incidence matrix of ϕ_d, with columns recording the numbers of a's and b's in the images of a and b. For the completed count column

$$c(z)=\begin{pmatrix}|z|_a+1\\ |z|_b+1\end{pmatrix},$$

one has

$$c(F_d(z))=L_dc(z), \qquad L_a=\begin{pmatrix}1&1\\0&1\end{pmatrix}, \quad L_b=\begin{pmatrix}1&0\\1&1\end{pmatrix}. \tag{4}$$

In particular, the greatest common divisor of the two completed counts is unchanged by F_d, and every central word has primitive completed counts.

Proof. The identities hold for the empty directive. Write $u=\phi_d(a)$ and $v=\phi_d(b)$. Appending a changes (u,v) to (u,vu) and g_d to g_du; appending b changes them to (uv,v) and g_dv. If $uv=abg_d$ and $vu=bag_d$, these updates immediately preserve both identities in (3). Counting their letters gives $c(g_d)=L_d(1,1)^{\mathsf T}$. Counting $F_d(z)=g_d\phi_d(z)$ then proves (4). Each L_d is an integral unimodular matrix. Such a matrix preserves the ideal generated by the entries of a column, and therefore their greatest common divisor. Apply this to $c(\varepsilon)=(1,1)^{\mathsf T}$ for the last assertion. □

Primitivity is only a necessary condition for centrality among palindromes. We shall use precisely that implication, never its converse. The maps F_d are injective: each of the two morphisms in (1) has a prefix code as its set of letter images, so is injective; a fixed prefix g_d can then be cancelled.

2.2. **A positive integral Gram model.** The Cohn representation used in this paper is

$$\mu(a)=A=\begin{pmatrix}2&1\\1&1\end{pmatrix}, \qquad \mu(b)=B=\begin{pmatrix}5&2\\2&1\end{pmatrix}, \qquad P=\begin{pmatrix}2&1\\1&0\end{pmatrix}, \quad e=\begin{pmatrix}2\\1\end{pmatrix}. \tag{5}$$

The matrices A,B are symmetric, positive definite and unimodular. Consequently $\mu(\widetilde{z})=\mu(z)^{\mathsf T}$, and for each palindrome z the matrix $\mu(z)$ is positive definite. Indeed, if $z=\widetilde{w}w$, it is $\mu(w)^{\mathsf T}\mu(w)$; if $z=\widetilde{w}cw$, it is $\mu(w)^{\mathsf T}\mu(c)\mu(w)$.

Define its *Gram label* $M(z)$ and *Gram root* $\rho(z)$ by

$$\mathcal{G}(z) = P\mu(z)P = \begin{pmatrix} M(z) & \rho(z) \\ \rho(z) & H(z) \end{pmatrix}. \tag{6}$$

These are positive integers, and

$$M(z)H(z) - \rho(z)^2 = 1. \tag{7}$$

In particular, $\rho(z)^2 \equiv -1 \pmod{M(z)}$. The label is also a primitive sum of two squares. To see this directly, put $P_1 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $P_2 = P$. Then $A = P_1^2$, $B = P_2^2$, and the factorizations above give $\mathcal{G}(z) = T^{\mathsf{T}}T$ with an integral unimodular T: use $T = \mu(w)P$ in the even case and $T = P_c\mu(w)P$ in the odd case, where $P_a = P_1$ and $P_b = P_2$. The first column of T is positive and primitive, so its squared Euclidean norm is $M(z)$.

There is also a direct connection with the more familiar trace label. For $J_* = AP^{-1} = \left(\begin{smallmatrix} 1 & 0 \\ 1 & -1 \end{smallmatrix}\right)$, one has

$$\mu(azb) = J_*\mathcal{G}(z)P, \qquad \operatorname{tr}\mu(azb) = 3M(z). \tag{8}$$

Both formulas follow by multiplication, using $B = P^2$. If z is central, azb is a proper lower Christoffel word, and the classical Cohn correspondence identifies $M(z)$ as a Markov number [12, Theorem 1]. Here a Markov number is an entry of a positive integral solution of $x^2 + y^2 + z^2 = 3xyz$. For a general palindrome, (8) remains an exact identity, but it does not assert that the label is a Markov number.

Among palindromes with prescribed completed counts, the central word is distinguished by the label itself. Lagisquet, Pelantová, Tavenas and Vuillon prove a minimum for the upper-right path value at fixed counts, with its exact equality cases [8, Proposition 21]; transported to the present Gram normalization, it says that for coprime completed counts every palindrome satisfies $M(z) \geqslant M(z_0)$, where z_0 is the central word with those counts, and that equality holds only for z_0. We record this because it fixes the direction of the comparison used below: a competing palindrome with the same counts can only have a larger label, never a smaller one. The statement is theirs; nothing in this paper strengthens it, and the noncoprime case is not used.

2.3. **What the Gram root counts.** The identity (7) makes $\rho(z)$ a square root of -1 modulo the label. That root also has a word-theoretic meaning, and the incidence transport of (4) is enough to find it. Write $\widehat{d}$ for the *reversal complement* of a word d, obtained by reading d backwards and exchanging the two letters.

Lemma 2.2 (The reversal complement transposes the incidence matrix)**.** *For every directive d, $L_{\widehat{d}} = L_d^{\mathsf{T}}$.*

Proof. The two matrices of (4) satisfy $L_a^{\mathsf{T}} = L_b$ and $L_b^{\mathsf{T}} = L_a$, so exchanging the letters of a directive transposes each factor of its incidence product. Reading the directive backwards reverses the order of those factors. Performing both operations therefore transposes the product. □

Proposition 2.3 (The doubled directive and its letter counts)**.** *Let d be a directive and let*

$$\begin{pmatrix} q \\ s \end{pmatrix} = c(\mathrm{Pal}(d)) = L_d \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

be its completed counts, so that $\gcd(q, s) = 1$. Put $m = q^2 + s^2$. Then the completed counts of the doubled central word $\mathrm{Pal}(\widehat{d}\,d)$ are

$$c(\mathrm{Pal}(\widehat{d}\,d)) = L_d^{\mathsf{T}} L_d \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \tag{9}$$

so the completed word $a\,\mathrm{Pal}(\widehat{d}\,d)b$ *has length exactly* m.[1] *Writing* r_d *for its number of letters* b*, one has* $0 < r_d < m$ *and*

$$r_d\, q \equiv s \pmod m, \qquad \textit{hence} \qquad r_d^2 \equiv -1 \pmod m. \tag{10}$$

The number of letters a *is* $m - r_d$.

Proof. By (2), $\mathrm{Pal}(\widehat{d}\,d) = F_{\widehat{d}}(\mathrm{Pal}(d))$, so (4) and lemma 2.2 give (9). Write $L_d = \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)$, so that $q = \alpha + \beta$ and $s = \gamma + \delta$, and $\alpha\delta - \beta\gamma = 1$. Expanding (9),

$$m - r_d = \alpha q + \gamma s, \qquad r_d = \beta q + \delta s,$$

whose sum is $(\alpha + \beta)^2 + (\gamma + \delta)^2 = q^2 + s^2$, which is the length assertion.

Both α and δ are at least one, since L_d has nonnegative entries and determinant one, and $\alpha = 0$ would force $-\beta\gamma = 1$. Hence $r_d \geqslant \delta s \geqslant 1$ and $m - r_d \geqslant \alpha q \geqslant 1$, which is the strict range.

Finally $\gcd(q, m) = \gcd(q, s^2) = 1$, so q is invertible modulo m, and $q^2 \equiv -s^2$. Therefore

$$r_d\, q = \beta q^2 + \delta s q \equiv -\beta s^2 + \delta s q = s\big(\delta(\alpha + \beta) - \beta(\gamma + \delta)\big) = s(\alpha\delta - \beta\gamma) = s \pmod m.$$

Squaring and cancelling the unit q^2 gives $r_d^2 \equiv -1$. □

The two letter counts of that word are not merely arithmetic data: they are also the two lengths into which it splits. Recall that a proper Christoffel word has a unique factorization into two Christoffel words, its *standard factorization*, and that the lengths of the two factors are the column sums of the incidence matrix of its directive [2, Corollaries 3.1–3.2].

Corollary 2.4 (The cut and the counts are the same pair)**.** *Let* d *be a directive, let* $\Omega = a\,\mathrm{Pal}(\widehat{d}\,d)b$*, and let* $\Omega = \Omega_1\Omega_2$ *be its standard factorization. Then*

$$|\Omega_1| = |\Omega|_a = m - r_d, \qquad |\Omega_2| = |\Omega|_b = r_d. \tag{11}$$

That is, Ω *is cut after exactly as many letters as it contains letters* a.

Proof. The directive of Ω is $\widehat{d}\,d$, so by lemma 2.2 its incidence matrix is $L_{\widehat{d}\,d} = L_{\widehat{d}} L_d = L_d^{\mathsf{T}} L_d$. This is a Gram matrix, hence symmetric, and therefore its column sums and its row sums are the same ordered pair:

$$\mathbf{1}^{\mathsf{T}} L_d^{\mathsf{T}} L_d = \big(L_d^{\mathsf{T}} L_d \mathbf{1}\big)^{\mathsf{T}}, \qquad \mathbf{1} = (1, 1)^{\mathsf{T}}.$$

The left side is the pair of factor lengths of the standard factorization, and the right side is, by (9), the pair of completed counts of Ω. The identification of the individual entries is the order of (9), and $|\Omega|_a = m - r_d$ was proved in proposition 2.3. □

Symmetry of a Gram matrix is the whole content: the cut point and the letter count are one pair of integers read along the two axes of a single matrix. Exchanging the two letters of d replaces that matrix by the Gram matrix of the rows of L_d, which has the same total mass and the two entries of (11) interchanged, so the cut is reflected in the middle of the word.

The Gram model sees the same root. Recall the factorization $\mathcal{G}(z) = T^{\mathsf{T}} T$ constructed above, with $T = \mu(w)P$ in the even case and $T = P_c\mu(w)P$ in the odd case.

[1] That a Markov number is a sum of two squares was obtained earlier, by a substitution argument on words resting on Justin's formula, by L. Vuillon and A. De Luca (personal communication, 2026). The proof below is independent of theirs and yields the letter counts as well as the length.

Corollary 2.5 (The Gram root as a letter count)**.** *Let z be a palindrome, let (q, s) be the first column of T, and put $\varepsilon_z = \det T$, which is $+1$ when $|z|$ is odd and -1 when $|z|$ is even. Then $M(z) = q^2 + s^2$ with $\gcd(q, s) = 1$, and*

$$\rho(z)\, q \equiv \varepsilon_z\, s \pmod{M(z)}.$$

Let d be the unique directive with completed counts (q, s). Then $\rho(z) = r_d$ when $\varepsilon_z = +1$, and $\rho(z) = M(z) - r_d$ when $\varepsilon_z = -1$.

Proof. Let (β, δ) be the second column of T. Then $M(z) = q^2 + s^2$ and $\rho(z) = \beta q + \delta s$ are the two entries of the first row of $T^{\mathsf{T}}T$, and $\det T = q\delta - s\beta = \varepsilon_z$. Since $\det \mu(w) = 1$, $\det P = \det P_1 = -1$, the two cases of the construction give $\varepsilon_z = -1$ and $\varepsilon_z = +1$ respectively. As in the proposition, $q^2 \equiv -s^2$ modulo $M(z)$, so

$$\rho(z)\, q = \beta q^2 + \delta s q \equiv s(q\delta - s\beta) = \varepsilon_z s.$$

The first column of T is positive and primitive, so (q, s) is the completed-count column of exactly one central word, namely $\mathrm{Pal}(d)$ for a unique directive d; this is the inverse of the unimodular transport (4). Both $\rho(z)$ and r_d lie strictly between 0 and $M(z)$, and by (10) they satisfy the same congruence up to the sign ε_z. A residue class modulo $M(z)$ meets that range once, so they agree when $\varepsilon_z = +1$ and are complementary when $\varepsilon_z = -1$. □

When z is central, azb is a proper lower Christoffel word and $M(z)$ is a Markov number, so $\rho(z)$ is then the classical root of -1 attached to the occurrence, and $q^2 + s^2$ is its primitive two-square representation. Proposition 2.3 says that this root is not an abstract residue: it is the number of letters b in one explicit Christoffel word of length $M(z)$, the one directed by the doubled directive $\widehat{d}\, d$. The statement itself needs no centrality, and its proof uses only (4) and a determinant. Corollary 2.4 adds that the same integer also measures where that word splits: the standard factorization cuts after $M(z) - \rho(z)$ letters when $\varepsilon_z = +1$, and after $\rho(z)$ letters when $\varepsilon_z = -1$. A root of -1 modulo a Markov number is thus visible twice in a single word, once as a count and once as a position.

2.4. **The Thue–Morse coding.** The label and the root have a second reading, in the coding of Markov numbers by Reutenauer and Vuillon [13], and a search for equal lengths among all palindromes of that coding finds the pairs of section 4. Let θ be the Thue–Morse morphism, $\theta(a) = ab$ and $\theta(b) = ba$, and recall the matrix $P_1 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ of the Gram factorization, with $P_1^2 = A$.

Proposition 2.6 (The Thue–Morse coding)**.** *For every word z,*

$$L_{\theta(F_a(z))} = P_1\, \mu(z)\, P_1. \tag{12}$$

Consequently, for every palindrome z the Christoffel word $\Omega_z = a\,\mathrm{Pal}(\theta(F_a(z)))b$ has length $M(z)$ and contains exactly $\rho(z)$ letters b. Moreover, F_a is a bijection from the set of palindromes onto the set of palindromes that begin with a and do not contain the factor bb.

Proof. The standard morphism of a concatenation of directives is the composition of their standard morphisms, so $L_{uv} = L_u L_v$; since θ is a morphism, it suffices to compare the images of the generators. On the word side, $L_{\theta(a)} = L_a L_b = A$ and

$$L_{\theta(\phi_a(b))} = L_{baab} = (L_b L_a)(L_a L_b) = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}.$$

On the matrix side, $P_1^{-1}AP_1 = A$ because $A = P_1^2$, and

$$P_1^{-1}BP_1 = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 7 & 5 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}.$$

Since $\theta(\phi_a(a)) = \theta(a)$, this proves $L_{\theta(\phi_a(c))} = P_1^{-1}\mu(c)P_1$ for both letters c. The product over the letters of z telescopes to $L_{\theta(\phi_a(z))} = P_1^{-1}\mu(z)P_1$, and therefore $L_{\theta(F_a(z))} = L_{\theta(a)}L_{\theta(\phi_a(z))} = P_1^2P_1^{-1}\mu(z)P_1$, which is (12).

Every directive D satisfies $c(\mathrm{Pal}(D)) = L_D(1,1)^{\mathsf{T}}$, by (2) and (4) applied to the empty word. Take $D = \theta(F_a(z))$ and write $\mu(z)e = (x,y)^{\mathsf{T}}$. Since $P_1(1,1)^{\mathsf{T}} = e$, the identity (12) gives

$$c(\mathrm{Pal}(D)) = P_1\mu(z)e = \begin{pmatrix} x+y \\ x \end{pmatrix}.$$

For a palindrome z one has $M(z) = e^{\mathsf{T}}\mu(z)e = 2x + y$, because the first column of P is e, and $\rho(z) = (1,0)\,\mu(z)e = x$, because the second row of P is $(1,0)$. The completed counts of $\mathrm{Pal}(D)$ are the letter counts of Ω_z, which are therefore $M(z) - \rho(z)$ and $\rho(z)$. The word Ω_z is a Christoffel word because $\mathrm{Pal}(D)$ is central.

For the last assertion, recall $F_a(z) = a\phi_a(z) = \psi_a(z)a$. The reversal of $\phi_a(z)$ is $\psi_a(\widetilde{z})$, so the reversal of $F_a(z)$ is $F_a(\widetilde{z})$. If z is a palindrome, $F_a(z)$ is thus a palindrome; it begins with a, and it has no factor bb because every b of $\phi_a(z)$ is followed by an a. Conversely, let $u = au'$ be a palindrome without the factor bb. The word u' is empty or ends with a, so each of its letters b is followed by the letter a, and u' factors uniquely over the code $\{a, ba\}$: $u' = \phi_a(z)$ and $u = F_a(z)$. Finally $F_a(z) = u = F_a(\widetilde{z})$, and the injectivity of F_a gives $z = \widetilde{z}$. □

For a central word $z = \mathrm{Pal}(v)$ one has $F_a(z) = \mathrm{Pal}(av)$ by (2). Then Ω_z is the word $a\,\mathrm{Pal}(\theta(\mathrm{Pal}(av)))b$ of Reutenauer and Vuillon, and the length statement is their theorem that $2 + |\mathrm{Pal}(\theta(\mathrm{Pal}(av)))|$ is the Markov number of the occurrence [13, Theorem 1]. The proposition adds the root. By (8) the first row of the Cohn matrix $\mu(azb) = J_*\mathcal{G}(z)P$ is $(M,\rho)P = (2M+\rho,\ M)$, so the number of letters b of their word is $\mu(azb)_{11} - 2M$, for every occurrence and with no choice of sign. For an arbitrary palindrome, $F_a(z)$ is an arbitrary palindrome that begins with a and avoids bb, and a coincidence of the lengths $|\mathrm{Pal}(\theta(F_a(z)))|$ is exactly an equality of Gram labels. The pair of the introduction is the first one: with $F_a(baab) = abaaaba$ and $F_a(abba) = aababaa$, the directives $\theta(abaaaba)$ and $\theta(aababaa)$ give central words whose completed counts are $(663, 467)$ and $(693, 437)$, and both completed words have length 1130.

The Thue–Morse directive is the doubled directive of proposition 2.3, which explains the sign in corollary 2.5.

Corollary 2.7 (The Thue–Morse directive is a doubled directive)**.** *Let z be a palindrome and let d be the directive of corollary 2.5, whose completed counts are the first column of T. Then $\theta(F_a(z)) = \widehat{d}\,d$ when $|z|$ is odd, and $\theta(F_a(z))$ is obtained from $\widehat{d}\,d$ by exchanging the two letters when $|z|$ is even. In particular the sign ε_z of corollary 2.5 is the exchange of letters, and the Gram root is the number of letters b of Ω_z in both parities.*

Proof. Every matrix of $\mathrm{SL}_2(\mathbb{Z})$ with nonnegative entries is a product of L_a and L_b in exactly one way, and the column $L_{d'}(1,1)^{\mathsf{T}}$ determines the directive d', as in the proof of corollary 2.5. In the odd case, $z = \widetilde{w}cw$, the matrix $L = P_c\mu(w)P_1$ has determinant $(-1)\cdot(-1) = 1$ and nonnegative entries, and $L(1,1)^{\mathsf{T}} = P_c\mu(w)e = Te_1$. Hence $L = L_d$, and by lemma 2.2

$$L_{\widehat{d}\,d} = L_d^{\mathsf{T}}L_d = P_1\mu(\widetilde{w})P_c^2\mu(w)P_1 = P_1\mu(z)P_1,$$

because P_c is symmetric and $P_c^2 = \mu(c)$. In the even case, $z = \widetilde{w}w$, the matrix $L = \mu(w)L_a$ has determinant one and nonnegative entries, and $L(1,1)^{\mathsf{T}} = \mu(w)e = Te_1$; hence $L = L_d$ and $L_{\widehat{d}\,d} = L_b\mu(\widetilde{w})\mu(w)L_a = L_b\mu(z)L_a$. With $S = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ one has $P_1 = L_aS = SL_b$, so $P_1\mu(z)P_1 = S\,L_{\widehat{d}\,d}\,S$, and conjugation by S exchanges L_a and L_b, that is, the two letters of a directive. In both cases (12) and the uniqueness of the factorization prove the claim. Exchanging the letters of a directive exchanges the two letter counts of its central word, which gives the statement about ε_z by comparison with corollary 2.5. □

The coding turns the question of this article into a question about the lengths of central words. The pairs classified in section 4 are coincidences of these lengths, and section 6 asks whether they are the only ones.

2.5. **The reflection attached to an ancestor.** Fix a directive d and abbreviate

$$G = \mu(g_d), \quad \nu_d = \mu \circ \phi_d, \quad M_0 = e^{\mathsf{T}}Ge, \quad \rho_0 = (Ge)_1, \quad r_0 = \rho_0/M_0. \tag{13}$$

The entries of Ge are positive, so $0 < r_0 < 1/2$. Equation (2) becomes

$$\mathcal{G}(F_d(z)) = PG\nu_d(z)P. \tag{14}$$

In particular, $G\nu_d(z)$ is symmetric for a palindrome z, although $\nu_d(z)$ itself need not be symmetric.

The ancestor determines the rational matrix

$$R_d = \frac{2ee^{\mathsf{T}}G}{M_0} - I. \tag{15}$$

It fixes e, squares to I, and satisfies $R_d^{\mathsf{T}}G = GR_d$ and $R_d^{\mathsf{T}}GR_d = G$. Thus it is an orthogonal reflection for the inner product defined by G. For example, if $G = T^{\mathsf{T}}T$, then TR_dT^{-1} is the ordinary Euclidean reflection in the line spanned by Te. In the Gram coordinates its form is particularly simple:

$$R_d = PJ_0P^{-1}, \qquad J_0 = \begin{pmatrix} 1 & 2r_0 \\ 0 & -1 \end{pmatrix}. \tag{16}$$

The next section determines exactly when this reflection carries a finite Cohn word column to another such column.

3. Decoding a rational reflection on finite words

The matrix identities for a standard directive supply many pairs related by its reflection: one may exchange the blocks ab and ba in any concatenation. The question is whether there are other pairs. We prove that there are none, even among arbitrary finite words. The proof reads two letters at a time. A change of projective coordinate places all possible first letters in two fixed intervals, and the final letter of the directive determines which inequalities exclude a wrong continuation. Section 4.3 proves the same statement for the reflection of every palindrome by a shorter argument in the slope coordinate. The cylinders of this section are kept because they also measure how far a reflected column can move, which is what section 5 needs.

For a directive d, retain the notation

$$U = \mu(\phi_d(a)), \qquad V = \mu(\phi_d(b)), \qquad G = \mu(\mathrm{Pal}(d)), \qquad \nu_d(w) = \mu(\phi_d(w)).$$

Thus ν_d evaluates a word by replacing a, b with U, V. Write $M_0 = e^{\mathsf{T}}Ge$, $\rho_0 = (Ge)_1$ and

$$K = 2ee^{\mathsf{T}}G - M_0I, \qquad R = K/M_0.$$

Since $(ee^{\mathsf{T}}G)^2 = M_0ee^{\mathsf{T}}G$, one has $K^2 = M_0^2I$; in particular R is an invertible involution. For a word x in the two blocks ab, ba, let $\widehat{x}$ exchange these blocks in their original order. The parsing into blocks is unambiguous because both blocks have length two. For

example, $\widehat{abba} = baab$. For a nonzero column v, the notation $[v]$ denotes its line in the real projective line.

Theorem 3.1 (Finite-word reflection decoder)**.** *For every finite directive d and every pair of finite words x, y, including the empty word,*

$$[R\nu_d(x)e] = [\nu_d(y)e] \quad \Longleftrightarrow \quad x \in \{ab, ba\}^*, \qquad y = \widehat{x}. \tag{17}$$

Whenever these conditions hold, the proportionality is exact:

$$R\nu_d(x)e = \nu_d(\widehat{x})e. \tag{18}$$

There is no palindrome or primitive-count hypothesis in this theorem. Those restrictions will enter only when the decoder is applied to the midpoint problem.

3.1. **The two block identities.** The literal standard identities proved in the preceding section give

$$UV = ABG, \qquad VU = BAG, \qquad G = G^{\mathsf{T}}. \tag{19}$$

Appending a to d replaces (U, V) by (U, VU), and appending b replaces it by (UV, V). The direction of this update will matter when we compare the two generators.

Direct multiplication gives $AB + BA = 6ee^{\mathsf{T}}$. For every symmetric matrix G one also has $\operatorname{tr}(ABG) = 3e^{\mathsf{T}}Ge$. Consequently

$$3K = UV + VU - \operatorname{tr}(UV)I, \qquad KUV = VUK, \qquad KVU = UVK, \qquad Ke = M_0 e. \tag{20}$$

Here the middle identities can be checked without a word argument: Cayley–Hamilton makes

$$(UV + VU - \operatorname{tr}(UV)I)UV = VUUV - I = VU(UV + VU - \operatorname{tr}(UV)I).$$

Interchanging U, V proves the other identity. In particular, these relations prove the forward construction in (18) by induction on the number of blocks. The rest of the section proves exhaustion.

3.2. **A common coordinate at the terminal slope.** Matrices act on slopes by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}(t) = \frac{at + b}{ct + d},$$

with the usual interpretation at infinity. The terminal column e has slope two. Set

$$\mathcal{J} = [8/5, 29/12], \qquad z = \frac{1}{t - 2}. \tag{21}$$

Conjugation by $P = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is exactly this change of coordinate, and sends the terminal slope to $z = \infty$. The usefulness of the coordinate is that every positive word already lies in the same interval:

$$A(\mathcal{J}) = [21/13, 70/41] \subset \operatorname{int} \mathcal{J}, \qquad B(\mathcal{J}) = [50/21, 169/70] \subset \operatorname{int} \mathcal{J}.$$

Both matrices have determinant one, so both maps are increasing on $\mathcal{J}$ and these endpoint evaluations prove the inclusions. The second is the tight one: its upper endpoint clears that of $\mathcal{J}$ by $29/12 - 169/70 = 1/420$, so the inclusion must be read in exact arithmetic. Since $2 \in \mathcal{J}$, induction applies to every finite positive word. The interval $\mathcal{J}$ is a rational neighbourhood of the interval between the attracting fixed points of A and B on slopes, the golden ratio $(1 + \sqrt{5})/2$ and $1 + \sqrt{2}$; its endpoints $8/5$ and $29/12$ are a Fibonacci and a Pell quotient just outside them. Moreover $A(t) < 2 < B(t)$ for every positive t, so the first letter of a nonempty word decides on which side of the terminal slope its column lies. In the coordinate z the two halves $[8/5, 2)$ and $(2, 29/12]$ of $\mathcal{J}$ become the rays $z \leqslant -5/2$ and $z \geqslant 12/5$, and the terminal slope 2 becomes $z = \infty$.

There are positive integers p, q such that

$$\operatorname{tr} U = 3p, \quad (P^{-1}UP)_{21} = -p, \qquad \operatorname{tr} V = 3q, \quad (P^{-1}VP)_{21} = q. \tag{22}$$

At the empty directive this holds with $(p, q) = (1, 2)$. For the induction step, multiplication with a general symmetric G gives

$$\operatorname{tr}(ABG) = \operatorname{tr}(BAG) = 3M_0, \qquad (P^{-1}ABGP)_{21} = -M_0, \qquad (P^{-1}BAGP)_{21} = M_0.$$

Thus the two directive updates replace q or p, respectively, by the positive integer M_0.

Define the positive reciprocal parameters and two coordinate centers by

$$\alpha_0 = p^{-2}, \qquad \beta_0 = q^{-2}, \qquad u = \frac{(P^{-1}UP)_{11}}{p}, \qquad v = \frac{(P^{-1}VP)_{11}}{q}.$$

Trace and determinant determine the remaining entries:

$$\begin{aligned} P^{-1}UP &= p \begin{pmatrix} u & u^2 - 3u + \alpha_0 \\ -1 & 3 - u \end{pmatrix}, \\ P^{-1}VP &= q \begin{pmatrix} v & 3v - v^2 - \beta_0 \\ 1 & 3 - v \end{pmatrix}. \end{aligned} \tag{23}$$

Their projective maps in the new coordinate are therefore

$$f(z) = -u - \frac{\alpha_0}{z + u - 3}, \qquad g(z) = v - \frac{\beta_0}{z + 3 - v}. \tag{24}$$

Lemma 3.2 (Uniform bounds and the last directive letter)**.** *For every directive,*

$$3 \leqslant u \leqslant 41/12, \qquad 5/2 \leqslant v \leqslant 21/8. \tag{25}$$

If $L = u + v - 3$, then

$$M_0 = pqL, \qquad \alpha_0 + \beta_0 = L(3 - L), \qquad 5/2 \leqslant L < 3. \tag{26}$$

In particular $p \geqslant 1$ and $q \geqslant 2$. For a nonempty directive the following alternatives hold:

last letter	*trace dominance*	*reciprocal bounds*
a	$q \geqslant 5p$	$\beta_0 \leqslant \alpha_0/25, \quad \alpha_0 \leqslant 1$
b	$p \geqslant (5/2)q$	$\alpha_0 \leqslant 4\beta_0/25, \quad \beta_0 \leqslant 1/4.$

(27)

Proof. The word $\phi_d(a)$ is either a or begins with ab; the word $\phi_d(b)$ is either b or begins with ba. This follows by induction from the two updates following (19). Consequently $U(2) = A(s)$, where $s = 2$ if $U = A$ and otherwise s is the slope of a column of a word beginning with b, so that $s \in B(\mathcal{J}) \subset [2, 29/12]$. Likewise $V(2) = B(s')$ with $s' = 2$ or $s' \in A(\mathcal{J}) \subset [8/5, 2]$. Since A and B are increasing,

$$5/3 = A(2) \leqslant U(2) \leqslant A(29/12) = 70/41, \qquad 50/21 = B(8/5) \leqslant V(2) \leqslant B(2) = 12/5.$$

On the other hand (23) gives $U(2) = 2 - 1/u$ and $V(2) = 2 + 1/v$: the column e has cusp coordinate ∞, and $f(\infty) = -u$, $g(\infty) = v$ by (24). Solving, $12/41 \leqslant 1/u \leqslant 1/3$ and $8/21 \leqslant 1/v \leqslant 2/5$, which is (25).

The lower-left entry of $P^{-1}VUP$ is $pq(u + v - 3)$, so it equals $M_0 = pqL$. The matrix $P^{-1}(UV + VU)P = 6P^{-1}ee^{\mathsf{T}}GP$ has zero second row. Its lower-right entry, expanded with (23), gives $\alpha_0 + \beta_0 = L(3 - L)$. The bounds on u, v give $L \geqslant 5/2$; positivity of $\alpha_0 + \beta_0$ then gives $L < 3$.

This argument has not required lower bounds on p, q beyond their positive integrality. It now shows that either update to pqL strictly increases the replaced coordinate. Starting at $(1, 2)$, we obtain $p \geqslant 1, q \geqslant 2$ at every stage. For an appended a, the new q is $L_{\text{old}}pq_{\text{old}} \geqslant 5p$. For an appended b, the new p is $L_{\text{old}}p_{\text{old}}q \geqslant (5/2)q$. Taking reciprocal squares gives (27). □

3.3. **First-letter intervals and the reflection.** Let $\mathcal{C}$ be the set of cusp slopes of all $\nu_d(w)e$. The empty word contributes infinity, and

$$\mathcal{C} = \{\infty\} \cup f(\mathcal{C}) \cup g(\mathcal{C}) \subset \mathcal{H} := \{z \leqslant -5/2\} \cup \{z \geqslant 12/5\} \cup \{\infty\}. \tag{28}$$

The inclusion follows by transforming $\mathcal{J}$ under (21), as explained above. Each of the maps (24) has a single pole: f at $z = 3 - u$ and g at $z = v - 3$. By (25) the first lies in $[-5/12, 0]$ and the second in $[-1/2, -3/8]$, so both poles sit in the gap between the two rays of $\mathcal{H}$, and their distances to the rays are bounded below as in table 1. On the positive ray the denominators $z + u - 3$ and $z + 3 - v$ are these distances, and on the negative ray they are their negatives; in particular no denominator vanishes on $\mathcal{H}$.

TABLE 1. The poles of the two first-letter maps (24) and lower bounds for their distances to the two rays of $\mathcal{H}$, from (25). Dividing the numerators α_0 and β_0 by these distances gives the distances from the centres $-u$ and v to the four endpoints of the cylinders (29).

map	pole	distance to $z \geqslant 12/5$	distance to $z \leqslant -5/2$
f (letter a)	$3 - u \in [-5/12,\ 0]$	$\geqslant 12/5 - 0 = 12/5$	$\geqslant -5/12 + 5/2 = 25/12$
g (letter b)	$v - 3 \in [-1/2,\ -3/8]$	$\geqslant 12/5 + 3/8 = 111/40$	$\geqslant -1/2 + 5/2 = 2$

A map $z \mapsto c - \beta/(z - z_0)$ with $\beta > 0$ sends every point at distance at least δ from its pole z_0 to within β/δ of c, below c to the right of the pole and above c to the left of it, and it sends ∞ to c. For f the centre is $c = -u$, the numerator α_0 and the pole $3 - u$; for g they are v, β_0 and $v - 3$. Applied to the four entries of table 1, this gives

$$\begin{aligned} f(\mathcal{H}) &\subset \mathcal{U} := [-u - 5\alpha_0/12, -u + 12\alpha_0/25], \\ g(\mathcal{H}) &\subset \mathcal{V} := [v - 40\beta_0/111, v + \beta_0/2]. \end{aligned} \tag{29}$$

For example, the lower end of $\mathcal{V}$ is $v - \beta_0 \cdot 40/111$: the point of the positive ray closest to the pole of g is $12/5$, at distance $12/5 - (v - 3) \geqslant 111/40$. Figure 1 shows the configuration at the empty directive.

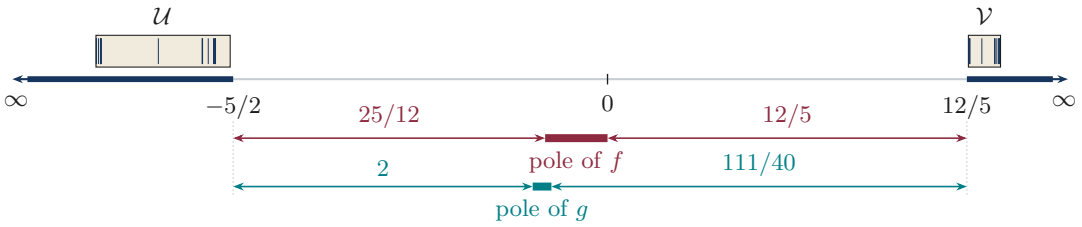


FIGURE 1. The cusp coordinate z at the empty directive, where $f(z) = -3 - 1/z$ and $g(z) = 5/2 - 1/(4z + 2)$. Every column of a nonempty word lies on one of the two rays of $\mathcal{H}$ (thick), and the terminal column is the point at infinity. The poles of f and g lie in the gap between the rays; the bars below the axis show the intervals $[-5/12, 0]$ and $[-1/2, -3/8]$ that contain them for every directive, and the arrows show the four distances of table 1, measured from the ends of these ranges. The shaded boxes are the cylinders $\mathcal{U}$ and $\mathcal{V}$ of (29), and the ticks above the axis are the columns of all words of length at most nine.

These two finite intervals lie strictly in the negative and positive components of $\mathcal{H}$:

$$-u + 12\alpha_0/25 \leqslant -3 + 12/25 < -5/2, \qquad v - 40\beta_0/111 \geqslant 5/2 - 10/111 > 12/5.$$

Consequently a finite slope in $\mathcal{C}$ determines the first letter, and inversion of its projective generator determines the remainder. The terminal value cannot belong to either finite interval. Induction therefore proves injectivity of finite-word evaluation at the terminal slope.

Put $\tau = 2\rho_0/M_0$. Since both entries of Ge are positive, $M_0 = 2\rho_0 + (Ge)_2$ gives $0 < \tau < 1$. Direct conjugation and (20) give

$$P^{-1}RP = \begin{pmatrix} 1 & \tau \\ 0 & -1 \end{pmatrix}, \qquad k(z) = -z - \tau, \qquad \tau = u - v + \delta, \quad \delta = \frac{\alpha_0 - \beta_0}{L}. \tag{30}$$

For the last equality, expand the upper-right entry of $3P^{-1}KP$ using (23). The map k fixes infinity and sends the two finite components of $\mathcal{H}$ to opposite signs. It need not preserve $\mathcal{H}$; whenever both endpoints belong to $\mathcal{C}$, their first letters must nevertheless be opposite.

We will also use positivity after an intermediate reflection, even when that reflected point has not yet been shown to be in the orbit. In the original coordinate,

$$R(t) = 2 - \frac{t-2}{1+\tau(t-2)} > 0 \qquad (t \in \mathcal{J}). \tag{31}$$

Indeed the denominator exceeds $3/5$. If $t < 2$, the value exceeds two; if $t \geqslant 2$, it is at least $2 - (t-2) \geqslant 19/12$. As U begins with A and V with B, they send every positive input below and above two, respectively. Hence fk has negative cusp slope on $\mathcal{C}$, while gk has positive cusp slope. These statements include the terminal input.

3.4. **Excluding a wrong second letter.** After opposite first letters have been removed, the two residual maps are

$$\begin{aligned} \mathcal{E}(z) &:= g^{-1}kf(z) = \frac{\beta_0}{\delta - \alpha_0/(z+u-3)} + v - 3, \\ \mathcal{D}(z) &:= f^{-1}kg(z) = \frac{\alpha_0}{\delta - \beta_0/(z+3-v)} + 3 - u. \end{aligned} \tag{32}$$

These formulas follow by solving (24) for its input. The identities (20) imply

$$\mathcal{E}g = fk, \qquad \mathcal{D}f = gk. \tag{33}$$

It remains to exclude an empty remainder or the wrong next letter. We prove precisely

$$\mathcal{E}(\{\infty\} \cup \mathcal{U}) \cap \mathcal{C} = \varnothing, \qquad \mathcal{D}(\{\infty\} \cup \mathcal{V}) \cap \mathcal{C} = \varnothing. \tag{34}$$

Only these domains are needed. In particular, no estimate across a pole of either residual map will be used.

A directive ending in a. Here $\beta_0 \leqslant \alpha_0/25$ and $\alpha_0 \leqslant 1$. Since $L < 3$,

$$\delta > \frac{8\alpha_0}{25} > 0, \qquad 0 < \frac{\beta_0}{\delta} < \frac{1}{8}.$$

It follows from $-1/2 \leqslant v - 3 \leqslant -3/8$ that $\mathcal{E}(\infty) \in (-1/2, -1/4)$. For $z \in \mathcal{U}$, the quantity $z + u - 3$ is negative, so the denominator of $\mathcal{E}$ is larger than δ. Therefore

$$\mathcal{E}(\{\infty\} \cup \mathcal{U}) \subset (-1/2, -1/4), \tag{35}$$

which is disjoint from $\mathcal{H}$.

For the other residual, direct simplification gives

$$\mathcal{D}(\infty) - v = \frac{L\beta_0}{\alpha_0 - \beta_0} > \frac{5}{2}\beta_0 > \frac{\beta_0}{2}.$$

If $z \in \mathcal{V}$, put $s = z + 3 - v$. The cylinder bound gives $s \geqslant 3 - 40\beta_0/111 > 2$. Its residual denominator h_0 satisfies

$$h_0 := \delta - \frac{\beta_0}{s} > \left(\frac{8}{25} - \frac{1}{50}\right)\alpha_0 = \frac{3\alpha_0}{10} > 0, \qquad h_0 < \delta < \frac{\alpha_0}{L}.$$

Using $L\delta = \alpha_0 - \beta_0$ now yields

$$\mathcal{D}(z) - v = \frac{\beta_0(1 + L/s)}{h_0} > \frac{L\beta_0}{\alpha_0} \geqslant \frac{5}{2}\beta_0 > \frac{\beta_0}{2}. \tag{36}$$

Both the terminal value and these finite values are positive and strictly above the entire positive cylinder $\mathcal{V}$. They cannot be in $\mathcal{C}$.

A directive ending in b. Now $\alpha_0 \leqslant 4\beta_0/25$ and $\beta_0 \leqslant 1/4$. Set $\eta = -\delta = (\beta_0 - \alpha_0)/L$. Then

$$\eta > \frac{7\beta_0}{25} > 0, \qquad 0 < \frac{\alpha_0}{\eta} < \frac{4}{7}.$$

At the terminal input,

$$\mathcal{E}(\infty) + u = -\frac{L\alpha_0}{\beta_0 - \alpha_0} < -10\alpha_0.$$

For $z \in \mathcal{U}$, put $s = 3 - z - u$. Since $\alpha_0 \leqslant 1/25$, the cylinder bound gives $s \geqslant 3 - 12\alpha_0/25 > 2$. The positive denominator

$$h_0 := \eta - \frac{\alpha_0}{s} > \left(\frac{7}{25} - \frac{2}{25}\right)\beta_0 = \frac{\beta_0}{5}, \qquad h_0 < \eta < \frac{\beta_0}{L},$$

allows the exact simplification

$$\mathcal{E}(z) + u = -\frac{\alpha_0(1 + L/s)}{h_0} < -\frac{L\alpha_0}{\beta_0} \leqslant -10\alpha_0. \tag{37}$$

These values and the terminal value are negative and strictly below $-u - 5\alpha_0/12$, the lower endpoint of $\mathcal{U}$. Thus they are outside $\mathcal{C}$.

Finally $\mathcal{D}(\infty) = -\alpha_0/\eta + 3 - u$. For $z \in \mathcal{V}$, the quantity $s = z + 3 - v$ is positive and

$$\mathcal{D}(z) = -\frac{\alpha_0}{\eta + \beta_0/s} + 3 - u.$$

The bounds $0 \leqslant u - 3 \leqslant 5/12$ and $0 < \alpha_0/\eta < 4/7$ give

$$\mathcal{D}(\{\infty\} \cup \mathcal{V}) \subset (-83/84, 0), \qquad \frac{5}{12} + \frac{4}{7} = \frac{83}{84}. \tag{38}$$

This interval is disjoint from $\mathcal{H}$. Equations (35)–(38) prove all of (34) for every nonempty directive.

3.5. **The empty directive and finite termination.** There is no last letter for the empty directive, so we check its residual exclusions directly in the original coordinate. Here $U = A$, $V = B$, $M_0 = 5$, and

$$K = \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}, \qquad E_0 = B^{-1}KA = \begin{pmatrix} 0 & 5 \\ 5 & -9 \end{pmatrix}, \qquad D_0 = A^{-1}KB = \begin{pmatrix} 9 & 5 \\ 5 & 0 \end{pmatrix}.$$

The use of K instead of R does not change a projective map. The terminal input is now two. We have $E_0(2) = 5 \notin \mathcal{J}$ and

$$E_0A(t) = \frac{5(t+1)}{t-4} < 0 \qquad (t \in \mathcal{J}).$$

Thus an empty source remainder or a second a is excluded. For the other residual,

$$D_0(2) = 23/10 \in (2, 50/21), \qquad D_0B(t) = \frac{55t + 23}{25t + 10}.$$

The latter map is decreasing on $\mathcal{J}$ and satisfies

$$2 < D_0 B(t) \leqslant 111/50 < 50/21.$$

All these values lie between the terminal slope and the least possible slope of a word starting with B. They lie outside the encoded orbit. This proves exactly the empty-directive version of (34), without a dominance assumption.

Proof of theorem 3.1. Represent (17) in cusp coordinates. If one word is empty, its slope is infinity; as $k(\infty) = \infty$ and no nonempty word has that slope, both words are empty. Otherwise their first letters are opposite.

Suppose $x = ax_1$ and $y = by_1$. Removing the target's first letter gives the residual relation $y_1(\infty) = \mathcal{E}(x_1(\infty))$, where words here denote their cusp maps. The first exclusion in (34) says that x_1 is neither empty nor starts with a. Hence $x_1 = bx_2$, and (33) gives

$$y_1(\infty) = fk(x_2(\infty)).$$

By (31) this is a finite negative cusp slope. Therefore $y_1 = ay_2$. Inverting f restores exactly the original relation $y_2(\infty) = k(x_2(\infty))$. Thus the first source block is ab and the first target block is ba.

If x begins with b, the second exclusion in (34) and the identity $\mathcal{D}f = gk$ give the exchanged conclusion: the source begins with ba, the target with ab, and removing those two letters restores the same reflection relation. The empty-directive calculation supplies the identical step in its original coordinate.

Each step removes two letters from each word. Induction on their total length terminates, and the terminal case forces the two words to end simultaneously. They are therefore precisely the block-exchanged pair in (17). Conversely, repeated use of (20) gives $K\nu_d(x) = \nu_d(\widehat{x})K$ for every block word. Applying both sides to e proves (18) and the converse implication. □

The exhaustion rests on the four strict residual exclusions and on the termination of the two-letter descent, both established above. The displayed identities and rational constants are also confirmed by exact computation in the supplementary material; no enumeration of directives or of finite words enters the proof.

4. Complete classification at the ancestor midpoint

The decoder concerns columns. The original arithmetic question concerns Gram matrices with the same first diagonal entry. The bridge between them is a full matrix conjugacy, which we prove before invoking the word theorem. This avoids any choice of a square root of a Gram matrix or of a middle letter in an odd palindrome.

Lemma 4.1 (The Gram bridge)**.** *Fix a directive d and two palindromes z_X, z_Y. Write $M_i = M(F_d(z_i))$ and $\rho_i = \rho(F_d(z_i))$. The following conditions are equivalent:*

$$M_X = M_Y = M, \qquad \rho_X + \rho_Y = 2r_0 M; \tag{39}$$

$$\nu_d(z_Y) = R_d \nu_d(z_X) R_d. \tag{40}$$

Under these conditions, $\nu_d(z_Y)e = R_d\nu_d(z_X)e$.

Proof. For a symmetric determinant-one matrix $\left(\begin{smallmatrix} M & \rho \\ \rho & H \end{smallmatrix}\right)$, conjugation by J_0 as a bilinear form gives

$$J_0^{\mathsf{T}} \begin{pmatrix} M & \rho \\ \rho & H \end{pmatrix} J_0 = \begin{pmatrix} M & 2r_0 M - \rho \\ 2r_0 M - \rho & 4r_0^2 M - 4r_0\rho + H \end{pmatrix}.$$

The lower-right entry is uniquely determined by the other entries and the determinant, because $M > 0$. Hence (39) is equivalent to $\mathcal{G}(F_d(z_Y)) = J_0^{\mathsf{T}}\mathcal{G}(F_d(z_X))J_0$. Substitute (14), multiply by P^{-1} on both sides, and use $R_d = PJ_0P^{-1}$ to obtain

$$G\nu_d(z_Y) = R_d^{\mathsf{T}}G\nu_d(z_X)R_d = GR_d\nu_d(z_X)R_d.$$

Cancel the invertible matrix G. All steps are reversible, proving the equivalence. Finally $R_d e = e$ gives the column identity. □

Theorem 4.2 (Palindromic pairs at the ancestor midpoint)**.** *Fix any finite directive d. Two binary palindromes z_X, z_Y satisfy (39) if and only if, for some word $w \in \{ab, ba\}^*$,*

$$z_X = \widetilde{w}w, \qquad z_Y = \widetilde{\widehat{w}}\,\widehat{w}. \tag{41}$$

The word w is uniquely determined by the ordered pair. If it contains k blocks, then

$$c(F_d(z_X)) = c(F_d(z_Y)) = (2k+1)L_d\begin{pmatrix}1\\1\end{pmatrix}. \tag{42}$$

The case $k = 0$ consists of two copies of the ancestor g_d. Every case $k > 0$ consists of two distinct, noncentral palindromes.

Proof. The Gram bridge and theorem 3.1, applied to the whole words z_X, z_Y, show that $z_X \in \{ab, ba\}^*$ and $z_Y = \widehat{z_X}$. Because z_X has even length and is a palindrome, each of its two letter counts is even. A word of m blocks ab, ba has m copies of each letter. Thus $m = 2k$ is even, and the midpoint of z_X falls between blocks. Its second half is a unique word $w \in \{ab, ba\}^*$ and $z_X = \widetilde{w}w$. Block exchange commutes with reversal: reversing a block interchanges ab and ba, and reversing a sequence reverses the order of its blocks. Consequently $z_Y = \widetilde{\widehat{w}}\,\widehat{w}$.

Conversely, the block intertwining identities in (20) imply $R_d\nu_d(x)R_d = \nu_d(\widehat{x})$ for every $x \in \{ab, ba\}^*$. Apply this with $x = \widetilde{w}w$ and then use the Gram bridge. This proves the converse at the matrix level, including the equality of labels.

Both preimages in (41) have $2k$ copies of each letter. Formula (4) proves (42). If $k > 0$, these completed counts have greatest common divisor $2k + 1 > 1$, whereas every central word has primitive completed counts. The words are distinct because block exchange changes their first letter, and F_d is injective. The empty case follows from $F_d(\varepsilon) = g_d$. □

Corollary 4.3 (Exclusion for central occurrences)**.** *Let d, t_X, t_Y be finite directives. If the central words $\mathrm{Pal}(dt_X)$ and $\mathrm{Pal}(dt_Y)$ have the same Gram label M and their roots sum to $2M\rho_0/M_0$, then $t_X = t_Y = \varepsilon$. In particular, a pair of distinct central words with equal labels cannot have its root midpoint at a common standard ancestor's root parameter.*

Proof. Use $z_i = \mathrm{Pal}(t_i)$ in the theorem. The images are central, so only $k = 0$ is possible. The palindromization recurrence strictly increases length at every nonempty directive step; hence $\mathrm{Pal}(t_i) = \varepsilon$ implies $t_i = \varepsilon$. □

4.1. **Exact examples and the role of primitivity.** At the empty directive, $G = I$, $(M_0, \rho_0) = (5, 2)$ and

$$R_\varepsilon = \frac{1}{5}\begin{pmatrix}3 & 4\\ 4 & -3\end{pmatrix}.$$

Take $w = ab$. Then $z_X = baab$ and $z_Y = abba$, and direct evaluation gives

$$\mathcal{G}(baab) = \begin{pmatrix}1130 & 467\\ 467 & 193\end{pmatrix}, \qquad \mathcal{G}(abba) = \begin{pmatrix}1130 & 437\\ 437 & 169\end{pmatrix}. \tag{43}$$

TABLE 2. Images of the fixed palindromes *baab* and *abba*. Here (M_0, ρ_0) belongs to the standard ancestor $\mathrm{Pal}(d)$, M is the common image label, and $\rho_- < \rho_+$ are the image roots. Every row satisfies $(\rho_- + \rho_+)/(2M) = \rho_0/M_0$. The completed counts have greatest common divisor three in every row.

d	M_0	ρ_0	M	ρ_-	ρ_+
ε	5	2	1130	437	467
a	13	5	19 786	7571	7649
b	29	12	219 530	90 753	90 927
ab	194	75	65 712 650	25 403 793	25 404 957
ba	433	179	730 645 066	302 043 659	302 046 257

Both determinants are one and $467 + 437 = 2 \cdot 1130 \cdot (2/5)$. This is a real equality of integral Gram labels. Its completed counts are $(3, 3)$, so the theorem explains exactly why it is irrelevant to a collision between distinct central occurrences.

The same preimages give a pair for every directive, and the whole table is one formula. It holds around every palindrome, standard or not.

Proposition 4.4 (The one-block pair in closed form)**.** *Let p be a palindrome with Gram label M_p and Gram root ρ_p. The words $p\,ba\,p\,ab\,p$ and $p\,ab\,p\,ba\,p$ have the common Gram label and the two Gram roots*

$$M = M_p(9M_p^2 + 1), \qquad \rho_\pm = (9M_p^2 + 1)\rho_p \pm 3M_p, \tag{44}$$

the sign being $+$ for $p\,ba\,p\,ab\,p$ and $-$ for $p\,ab\,p\,ba\,p$. In particular the two roots differ by exactly $6M_p$, and their mean is $M\rho_p/M_p$. For $p = \mathrm{Pal}(d)$ the two words are $F_d(baab)$ and $F_d(abba)$, and $(M_p, \rho_p) = (M_0, \rho_0)$.

Proof. Put $W = \mu(p)$ and $\mathcal{G}_p = PWP = \left(\begin{smallmatrix} m & r \\ r & h \end{smallmatrix}\right)$, a symmetric matrix with $mh - r^2 = 1$, where $m = M_p$ and $r = \rho_p$. Then $\mu(p\,ba\,p\,ab\,p) = W\,BA\,W\,AB\,W$, and inserting $P^{-1}P$ between the factors gives

$$\mathcal{G}(p\,ba\,p\,ab\,p) = \mathcal{G}_p\,K\,\mathcal{G}_p\,K^{\mathsf{T}}\mathcal{G}_p, \qquad K = P^{-1}BAP^{-1} = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix},$$

where $K^{\mathsf{T}} = P^{-1}ABP^{-1}$ because A, B and P are symmetric. A direct multiplication gives

$$K\mathcal{G}_pK^{\mathsf{T}} = \begin{pmatrix} 9m - 6r + h & 3m - r \\ 3m - r & m \end{pmatrix}, \qquad \mathcal{G}_pK\mathcal{G}_pK^{\mathsf{T}} = \begin{pmatrix} 9m^2 - 3mr + 1 & 3m^2 \\ 9mr - 3r^2 + 3 & 3mr + 1 \end{pmatrix},$$

where $mh = r^2 + 1$ was used in the second product. Multiplying by the first column $(m, r)^{\mathsf{T}}$ of $\mathcal{G}_p$, the first column of the Gram matrix is

$$\begin{pmatrix} (9m^2 - 3mr + 1)m + 3m^2r \\ (9mr - 3r^2 + 3)m + (3mr + 1)r \end{pmatrix} = \begin{pmatrix} m(9m^2 + 1) \\ (9m^2 + 1)r + 3m \end{pmatrix}.$$

The second word is obtained by exchanging the blocks, so its Gram matrix is

$$\mathcal{G}(p\,ab\,p\,ba\,p) = \mathcal{G}_p\,K^{\mathsf{T}}\mathcal{G}_p\,K\,\mathcal{G}_p,$$

and in the same way

$$K^{\mathsf{T}}\mathcal{G}_pK = \begin{pmatrix} 9m + 6r + h & -3m - r \\ -3m - r & m \end{pmatrix}, \qquad \mathcal{G}_pK^{\mathsf{T}}\mathcal{G}_pK = \begin{pmatrix} 9m^2 + 3mr + 1 & -3m^2 \\ 9mr + 3r^2 - 3 & 1 - 3mr \end{pmatrix},$$

and the first column of its Gram matrix is

$$\begin{pmatrix} (9m^2+3mr+1)m-3m^2r \\ (9mr+3r^2-3)m+(1-3mr)r \end{pmatrix} = \begin{pmatrix} m(9m^2+1) \\ (9m^2+1)r-3m \end{pmatrix}.$$

The mean of the two roots is $(9m^2+1)r = Mr/m$. For $p = \mathrm{Pal}(d)$, (3) gives $\phi_d(ba) = ba\,p$ and $\phi_d(ab) = ab\,p$, so that $F_d(baab) = p\,\phi_d(ba)\phi_d(ab) = p\,ba\,p\,ab\,p$, and likewise for $abba$. □

Every row of table 2 is (44) evaluated at the ancestor of that row. The entries were also obtained directly from the literal words, and the determinant, midpoint and count identities are confirmed by exact integer arithmetic in the supplementary material; the theorem, including its exhaustion statement, depends on neither.

The theorem uses neither an odd-label hypothesis nor a factorization of the common label. It also includes all three possible middle types of a palindrome: empty, a, and b. Their apparent difference disappears because the full-word decoder forces every reflected preimage in the midpoint case to have the even form (41). This is the main structural gain of passing from square-root factors to whole Gram matrices.

4.2. **Every palindrome is a centre.** Proposition 4.4 uses nothing about the centre beyond the symmetry and the determinant of its Gram matrix, and the same is true of the construction half of theorem 4.2. Fix a palindrome p and put

$$G = \mu(p), \quad M_p = e^{\mathsf{T}}Ge, \quad \rho_p = (Ge)_1, \quad K_p = 2ee^{\mathsf{T}}G - M_pI, \quad R_p = K_p/M_p.$$

For blocks $W_1, \dots, W_n \in \{ab, ba\}$ write

$$x_p = p\,W_1\,p\,W_2\,p\cdots p\,W_n\,p,$$

and let $\widehat{x}_p$ be the word obtained by exchanging every block. The word x_p is a palindrome exactly when $n = 2k$ is even and W_{2k+1-i} is the exchange of W_i for every i, since a middle block would have to equal its own exchange. We then say that x_p is *structured around* p, and $\widehat{x}_p$ is structured around p as well.

Proposition 4.5 (Block exchange around an arbitrary palindrome)**.** *For every palindrome p and all blocks $W_1, \dots, W_n \in \{ab, ba\}$,*

$$K_p\,\mu(W_1p\cdots W_np) = \mu(\widehat{W}_1p\cdots\widehat{W}_np)\,K_p,$$
$$R_p\,\mu(W_1p\cdots W_np)\,e = \mu(\widehat{W}_1p\cdots\widehat{W}_np)\,e.$$

Moreover $e^{\mathsf{T}}\mu(x_p)e = e^{\mathsf{T}}\mu(\widehat{x}_p)e$; for palindromic block words these are the Gram labels. If x_p is structured around p with $2k$ blocks and common label M, then $\mu(\widehat{x}_p) = R_p^{\mathsf{T}}\mu(x_p)R_p$, the roots satisfy $\rho(x_p) + \rho(\widehat{x}_p) = 2M\rho_p/M_p$, and both words have completed counts $(2k+1)\,c(p)$. For $k \geqslant 1$ the two words are distinct and neither is central.

Proof. The derivation of (20) uses only that G is symmetric, through $\mathrm{tr}(ABG) = 3e^{\mathsf{T}}Ge$, and that the products have determinant one, through the Cayley–Hamilton identity. With $G = \mu(p)$ it gives $K_p\,ABG = BAG\,K_p$ and $K_p\,BAG = ABG\,K_p$. Hence $K_p\Pi = \widehat{\Pi}K_p$ for $\Pi = \mu(W_1p)\cdots\mu(W_np)$ and its exchange $\widehat{\Pi}$, which is the displayed identity; applying it to e and using $K_pe = M_pe$ gives the column identity. Moreover $K_pe = 2e(e^{\mathsf{T}}Ge) - M_pe = M_pe$ and $e^{\mathsf{T}}GK_p = M_pe^{\mathsf{T}}G$. Since $\mu(x_p) = G\Pi$,

$$M_p\,e^{\mathsf{T}}\mu(\widehat{x}_p)e = e^{\mathsf{T}}G\widehat{\Pi}K_pe = e^{\mathsf{T}}GK_p\Pi e = M_p\,e^{\mathsf{T}}\mu(x_p)e.$$

Next, $(ee^{\mathsf{T}}G)^2 = M_pee^{\mathsf{T}}G$ gives $K_p^2 = M_p^2I$, so $\widehat{\Pi} = R_p\Pi R_p$; and $GR_pG^{-1} = R_p^{\mathsf{T}}$ because G is symmetric. Therefore $\mu(\widehat{x}_p) = GR_p\Pi R_p = R_p^{\mathsf{T}}\mu(x_p)R_p$. Since P is symmetric, this

gives $\mathcal{G}(\widehat{x}_p) = (P^{-1}R_pP)^{\mathsf{T}}\,\mathcal{G}(x_p)\,(P^{-1}R_pP)$. Since $P^{-1}e = e_1$ and $e^{\mathsf{T}}GP = e_1^{\mathsf{T}}PGP = (M_p, \rho_p)$,

$$P^{-1}R_pP = \frac{2e_1(M_p,\rho_p) - M_pI}{M_p} = \begin{pmatrix} 1 & 2\rho_p/M_p \\ 0 & -1 \end{pmatrix},$$

and the computation in the proof of lemma 4.1 shows that the congruence by this matrix takes a Gram matrix with first column (M, ρ) to one with first column $(M,\, 2M\rho_p/M_p - \rho)$. This is the root relation. The word x_p contains $2k+1$ copies of p and $2k$ blocks, each with one letter of each kind, which gives the completed counts. For $k \geqslant 1$ their greatest common divisor is at least $2k+1$, while central words have coprime completed counts by lemma 2.1; and the two words differ at the first letter of the first block. □

The exchange also has a transparent arithmetic. Recall that the roots r of -1 modulo a label M correspond bijectively to the primitive Gaussian integers $x + iy$ of norm M taken up to units, through $r \equiv xy^{-1} \pmod{M}$, and that replacing r by $-r$ replaces $x + iy$ by its conjugate.

Proposition 4.6 (The arithmetic of an exchange)**.** *Let x_p be structured around p with $2k \geqslant 2$ blocks and common label M. Then M_p divides M, the quotient $N = M/M_p$ is prime to M_p, and*

$$\rho(x_p) \equiv \rho(\widehat{x}_p) \equiv \rho_p \pmod{M_p}, \qquad \rho(x_p) + \rho(\widehat{x}_p) \equiv 0 \pmod{N}.$$

The exchange therefore keeps the Gaussian factor of norm M_p and conjugates the factor of norm N.

Proof. As in the proof of proposition 4.4, inserting $P^{-1}P$ between the factors gives $\mathcal{G}(x_p) = \mathcal{G}_pY_1\mathcal{G}_pY_2\cdots Y_n\mathcal{G}_p$, where $\mathcal{G}_p = \mathcal{G}(p)$ and Y_i is K when $W_i = ba$ and K^{T} when $W_i = ab$. Directly from these two matrices, $e_2^{\mathsf{T}}K = e_1^{\mathsf{T}}$, $e_2^{\mathsf{T}}K^{\mathsf{T}} = -e_1^{\mathsf{T}}$, $Ke_2 = -e_1$ and $K^{\mathsf{T}}e_2 = e_1$; and modulo M_p, entrywise, $e_1^{\mathsf{T}}\mathcal{G}_p \equiv \rho_p e_2^{\mathsf{T}}$ and $\mathcal{G}_pe_1 \equiv \rho_p e_2$, because the first row of $\mathcal{G}_p$ is (M_p, ρ_p). Reading $M(x_p) = e_1^{\mathsf{T}}\mathcal{G}(x_p)e_1$ from the left, each factor $\mathcal{G}_pY_i$ turns a multiple of e_1^{T} into $\pm\rho_p$ times e_1^{T}, so $M(x_p) \equiv \pm\rho_p^n\, e_1^{\mathsf{T}}\mathcal{G}_pe_1 = \pm\rho_p^n M_p \equiv 0$. Reading $\rho(x_p) = e_2^{\mathsf{T}}\mathcal{G}(x_p)e_1$ from the right, the column starts as ρ_pe_2, and each factor $\mathcal{G}_pY_i$ multiplies a multiple of e_2 by ρ_p and by -1 if $W_i = ba$, $+1$ if $W_i = ab$. Hence $\rho(x_p) \equiv \rho_p^{2k+1}(-1)^j$, where j is the number of blocks ba. For a structured word the blocks pair off as W_i and its exchange, so $j = k$; and $\rho_p^2 \equiv -1$ because $\det \mathcal{G}_p = 1$. Hence $\rho(x_p) \equiv (-1)^k(-1)^k\rho_p = \rho_p$, and the same holds for $\widehat{x}_p$.

For the coprimality write $\rho(x_p) = \rho_p + M_ps$ and $\rho(\widehat{x}_p) = \rho_p + M_ps'$. By proposition 4.5 their sum is $2\rho_pN$, so $M_p(s+s') = 2\rho_p(N-1)$. A prime dividing both M_p and N divides neither $N-1$ nor ρ_p, which is prime to M_p, so it must divide 2; then 4 divides $M = M_pN$. But M is a sum of two coprime squares, the squared norm of the first column of T, and such a sum is 1 or 2 modulo 4. Hence $\gcd(M_p, N) = 1$, and $\rho(x_p) + \rho(\widehat{x}_p) = 2\rho_pN \equiv 0 \pmod{N}$. By the Chinese remainder theorem the two roots agree modulo M_p and are opposite modulo N; a root and its negative select conjugate Gaussian primes, which gives the last assertion. □

With the centre $p = baab$, whose label is 1130 and root 467, the closed form (44) gives the label $1130 \cdot 11\,492\,101 = 12\,986\,074\,130$ and the roots $5\,366\,814\,557$ and $5\,366\,807\,777$. Since $baab$ is not central, this pair lies outside table 2, and the same label shows that the midpoint condition of theorem 4.2 cannot be removed.

Remark 4.7 (The midpoint hypothesis cannot be dropped)**.** Exactly four palindromes carry the label of the example above, $M = 12\,986\,074\,130$, namely

$$Z_1 = baab\,ba\,baab\,ab\,baab, \qquad Z_2 = baab\,ab\,baab\,ba\,baab,$$
$$Z_3 = abba\,ba\,abba\,ab\,abba, \qquad Z_4 = abba\,ab\,abba\,ba\,abba.$$

The pairs (Z_1, Z_2) and (Z_3, Z_4) are exchanges around *baab* and *abba*. Read in blocks of two letters, all four words lie in $\{ab, ba\}^*$, and (Z_1, Z_4) and (Z_2, Z_3) are exchanges around the empty word, with eight blocks each. The remaining pairs (Z_1, Z_3) and (Z_2, Z_4) have equal labels and lie in the image of the empty directive, but neither word is the block exchange of the other: their root midpoints are $2/5 \pm 3390/M$ instead of the ancestor ratio $2/5$. Without the midpoint hypothesis, the conclusion of theorem 4.2 fails. A label can also be carried by more than two palindromes, and two palindromes with the same label need not be exchanges of each other.

The four words form a single class under exchanges, each around its own centre, and the two pairs that are not exchanges have their midpoints shifted off every centre. The next subsection shows that a midpoint at the ratio of a palindrome forces an exchange around it, at every centre, and section 4.4 explains the shifted midpoints.

4.3. **Decoding at every palindromic centre.** Proposition 4.5 produces pairs of word columns related by the reflection R_p of any palindrome p. We now prove that there are no others, for every palindrome and all finite words, and deduce the classification of theorem 4.2 at every centre, without the hypotheses of a common standard image and of a common prefix. The argument is shorter than that of section 3. It works with the letters a and b in the slope coordinate, reads the first two letters of each word, and then reads the whole centre letter by letter through the transposed reflection. Beyond the symmetry of $\mu(p)$, the only number attached to the centre that enters the estimates is the window in which its ratio lies; the letters of the centre are read one at a time, by estimates valid for every word.

We write $t(v) = v_1/v_2$ for the slope of a column v, with $t(v) = \infty$ when $v_2 = 0$, and $t(w)$ for the slope of $\mu(w)e$; a matrix acts on slopes by fractional linear transformation. Two nonzero columns span the same line exactly when their slopes agree. By section 3, every word slope lies in $\mathcal{J} = [8/5, 29/12]$, with $t(\varepsilon) = 2$, $t(aw) \in A(\mathcal{J}) = [21/13, 70/41]$ and $t(bw) \in B(\mathcal{J}) = [50/21, 169/70]$, because $t(cw) = \mu(c)(t(w))$ and the two maps are increasing on $\mathcal{J}$. So $t(w) < 2$, $t(w) = 2$ or $t(w) > 2$ according as w begins with a, is empty, or begins with b, and no word slope is infinite, nonpositive, smaller than $21/13$ or larger than $169/70$. We call this the *first-letter rule*.

Fix a palindrome p and write $G = \mu(p)$, $M = M_p$, $\rho = \rho_p$, $K = K_p$ and $\tau = 2\rho/M$. Since $Ge = (\rho, M - 2\rho)^{\mathsf{T}}$ and $e^{\mathsf{T}}G = (Ge)^{\mathsf{T}}$,

$$K = \begin{pmatrix} 4\rho - M & 4M - 8\rho \\ 2\rho & M - 4\rho \end{pmatrix}, \qquad K^{\mathsf{T}} = \begin{pmatrix} 4\rho - M & 2\rho \\ 4M - 8\rho & M - 4\rho \end{pmatrix}. \tag{45}$$

Five facts follow by direct computation. First, $K^2 = M^2 I$, $Ke = Me$ and $GK = K^{\mathsf{T}}G$, as in the proof of proposition 4.5; hence $K^{\mathsf{T}}Ge = GKe = M\,Ge$, and $K^{\mathsf{T}}(1,-2)^{\mathsf{T}} = 2Ge\,e^{\mathsf{T}}(1,-2)^{\mathsf{T}} - M(1,-2)^{\mathsf{T}} = -M(1,-2)^{\mathsf{T}}$. Second, multiplying (45) by AB, BA and A,

$$KAB = BAK^{\mathsf{T}}, \qquad KBA = ABK^{\mathsf{T}}, \qquad B^{-1}KA = M \begin{pmatrix} 0 & 1 \\ 1 & -1 - \tau \end{pmatrix}; \tag{46}$$

for instance both upper-left entries in the first identity equal $12(4\rho - M) + 7(4M - 8\rho) = 16M - 8\rho$. Third, the slope $t(p) = \rho/(M - 2\rho)$ lies in $\mathcal{J}$ and $\tau = 2t(p)/(2t(p) + 1)$ increases with it, so $16/21 \leqslant \tau \leqslant 29/35$, the values at $8/5$ and at $29/12$. Fourth, for a real t, with $\eta = 1 + \tau(t - 2)$,

$$K \begin{pmatrix} t \\ 1 \end{pmatrix} = M \begin{pmatrix} 2\eta + 2 - t \\ \eta \end{pmatrix}, \qquad K^{\mathsf{T}} \begin{pmatrix} t \\ 1 \end{pmatrix} = M \begin{pmatrix} (2\tau - 1)t + \tau \\ (4 - 4\tau)t + 1 - 2\tau \end{pmatrix}. \tag{47}$$

Fifth, a relation $[T\mu(x)e] = [\mu(y)e]$ with T^2 a nonzero multiple of I is symmetric in x and y: applying T to both sides gives $[\mu(x)e] = [T\mu(y)e]$.

Lemma 4.8 (The first two letters)**.** *Let p be a palindrome and x, y words with $[K_p\mu(x)e] = [\mu(y)e]$. Then x is empty if and only if y is empty; if they are not empty, their first letters differ; and if x begins with a, then $x = ab\,x_1$ and $y = ba\,y_1$ for words x_1, y_1 with $[K_p^{\mathsf T}\mu(x_1)e] = [\mu(y_1)e]$.*

Proof. First letters. Let $t = t(x)$. Since $t - 2 \geqslant -2/5$ and $\tau \leqslant 29/35$, the number $\eta = 1 + \tau(t-2)$ is at least $1 - 58/175 = 117/175 > 0$. By (47) the column $K\mu(x)e$ is a positive multiple of $(2\eta + 2 - t, \eta)^{\mathsf T}$, so $t(y) = 2 + (2 - t(x))/\eta$, and $t(y) - 2$ has the sign of $2 - t(x)$, zero included. The first-letter rule gives the first two assertions.

The second letter of x. Let $x = ax'$, so that $y = by'$. Multiplying $[KA\mu(x')e] = [B\mu(y')e]$ by B^{-1} and using (46), the column $B^{-1}KA\mu(x')e$ is a positive multiple of $(1,\, t(x') - 1 - \tau)^{\mathsf T}$, so $t(y') = 1/(t(x') - 1 - \tau)$, read as ∞ when the denominator vanishes. If x' were empty, then $t(y') = 1/(1-\tau) \geqslant 21/5 > 169/70$. If x' began with a, then $t(x') \leqslant 70/41 < 37/21 \leqslant 1 + \tau$, and $t(y')$ would be negative. Neither is a word slope, so $x' = bx_1$.

The second letter of y. Now $[KAB\mu(x_1)e] = [B\mu(y')e]$, and (46) turns it into $[AK^{\mathsf T}\mu(x_1)e] = [\mu(y')e]$. By (47), with $t = t(x_1) \geqslant 8/5$, the column $K^{\mathsf T}\mu(x_1)e$ is a positive multiple of $((2\tau - 1)t + \tau,\, (4 - 4\tau)t + 1 - 2\tau)^{\mathsf T}$. Its first entry is positive because $\tau > 1/2$, and its second entry increases with t and is at least $(37 - 42\tau)/5 \geqslant 11/25$ at $t = 8/5$. So $K^{\mathsf T}\mu(x_1)e$ has a positive slope s, and $t(y') = A(s)$ lies strictly between 1 and 2. By the first-letter rule $y' = ay_1$, and cancelling A gives $[K^{\mathsf T}\mu(x_1)e] = [\mu(y_1)e]$. □

The transposed reflection $K^{\mathsf T}$ fixes the column Ge of the centre and reverses $(1, -2)^{\mathsf T}$. The next lemma shows that such a reflection can relate two word columns only if both words begin as the centre does.

Lemma 4.9 (One letter of a fixed word)**.** *Let T be a real matrix and $\kappa \neq 0$ with $Tu = \kappa u$ and $Tv = -\kappa v$, where $u = \mu(q)e$ for a nonempty word q with first letter c, and v is a column whose slope ω lies in $[-1, -2/5]$. If x and y are words with $[T\mu(x)e] = [\mu(y)e]$, then both begin with c.*

Proof. Put $\theta = t(q)$, which lies in $A(\mathcal J)$ if $c = a$ and in $B(\mathcal J)$ if $c = b$, and $\Omega = -\omega \in [2/5, 1]$. For a real $t \neq \omega$ let $\sigma(t) = (t - \theta)/(t - \omega)$. Writing $(t, 1)^{\mathsf T} = \alpha(\theta, 1)^{\mathsf T} + \beta(\omega, 1)^{\mathsf T}$ with $\alpha = (t - \omega)/(\theta - \omega)$ and $\beta = (\theta - t)/(\theta - \omega)$, one has $T(t, 1)^{\mathsf T} = \kappa(\alpha(\theta, 1)^{\mathsf T} - \beta(\omega, 1)^{\mathsf T})$, and when its slope t' is finite, $\sigma(t') = \beta/\alpha = -\sigma(t)$. The relation gives $t(y) = t'$ for $t = t(x)$, so $\sigma(t(y)) = -\sigma(t(x))$. On (ω, ∞), which contains every word slope, σ is increasing, with derivative $(\theta - \omega)/(t - \omega)^2$, and $\sigma < 1$.

The case $c = a$. Suppose that x does not begin with a, so $t(x) \geqslant 2$ and $\sigma(t(x)) \geqslant \sigma_0 = (2 - \theta)/(2 + \Omega) > 0$. Solving $\sigma(t(y)) = -\sigma(t(x))$ for $t(y)$ and using that $\sigma \mapsto \sigma/(1 + \sigma)$ increases,

$$t(y) = \theta - \frac{\sigma(t(x))}{1 + \sigma(t(x))}(\theta + \Omega) \leqslant \theta - \frac{\sigma_0}{1 + \sigma_0}(\theta + \Omega) = \theta - (2 - \theta)\,\frac{\theta + \Omega}{4 - \theta + \Omega}.$$

The last quotient increases with Ω, because $\theta < 2$, and with θ, so it is at least its value $131/181$ at $\theta = 21/13$, $\Omega = 2/5$. With $\theta \leqslant 70/41$ and $2 - \theta \geqslant 12/41$ this gives $t(y) \leqslant 70/41 - (12/41)(131/181) = 11098/7421 < 21/13$, which is the slope of no word.

The case $c = b$. Suppose that x does not begin with b, so $t(x) \leqslant 2 < \theta$ and $-\sigma(t(x)) \geqslant \sigma_0 = (\theta - 2)/(2 + \Omega) > 0$; also $\sigma_0 \leqslant 29/168 < 1$. Since $t(y)$ is a word slope, it lies in (ω, ∞), so $-\sigma(t(x)) = \sigma(t(y)) < 1$. Solving as before, and using that $\sigma \mapsto \sigma/(1 - \sigma)$ increases on $[0, 1)$,

$$t(y) \geqslant \theta + \frac{\sigma_0}{1 - \sigma_0}(\theta + \Omega) = \theta + (\theta - 2)\,\frac{\theta + \Omega}{4 - \theta + \Omega}.$$

The last quotient now decreases with Ω, because $\theta > 2$, and increases with θ, so it is at least its value $71/55$ at $\theta = 50/21$, $\Omega = 1$. With $\theta \geqslant 50/21$ and $\theta - 2 \geqslant 8/21$ this gives $t(y) \geqslant 50/21 + (8/21)(71/55) = 158/55 > 169/70$, which is the slope of no word.

In both cases x begins with c. The columns u and v have different slopes, so $T^2 = \kappa^2 I$, the relation is symmetric, and the same argument shows that y begins with c. □

Lemma 4.10 (Reading the centre)**.** *Let p be a palindrome. If two words x and y satisfy $[K_p^{\mathsf{T}}\mu(x)e] = [\mu(y)e]$, then $x = p\,x_0$ and $y = p\,y_0$, where the words x_0 and y_0 satisfy $[K_p\mu(x_0)e] = [\mu(y_0)e]$.*

Proof. Write $p = c_1 \cdots c_n$, and for $0 \leqslant j \leqslant n$ put $\Pi_j = \mu(c_1 \cdots c_j)$ and $T_j = \Pi_j^{-1} K^{\mathsf{T}} \Pi_j$. Then $T_0 = K^{\mathsf{T}}$ and $T_n = G^{-1}K^{\mathsf{T}}G = K$. For $j < n$ let $q_j = c_{j+1} \cdots c_n$, a nonempty word with first letter c_{j+1} and $\Pi_j \mu(q_j) = G$. Then $T_j\mu(q_j)e = \Pi_j^{-1}K^{\mathsf{T}}Ge = M\mu(q_j)e$, and the column $v_j = \Pi_j^{-1}(1,-2)^{\mathsf{T}}$ satisfies $T_j v_j = -Mv_j$. Its slope ω_j lies in $[-1,-2/5]$. Indeed $\omega_0 = -1/2$ and $\omega_j = \mu(c_j)^{-1}(\omega_{j-1})$; for $t < 0$,

$$A^{-1}(t) = \frac{t-1}{2-t} \in \Big(-1, -\frac{1}{2}\Big), \qquad B^{-1}(t) = \frac{t-2}{5-2t} \in \Big(-\frac{1}{2}, -\frac{2}{5}\Big),$$

because $A^{-1}(t) + 1 = 1/(2-t) > 0$, $A^{-1}(t) + 1/2 = t/(2(2-t)) < 0$, $B^{-1}(t) + 1/2 = 1/(2(5-2t)) > 0$ and $B^{-1}(t) + 2/5 = t/(5(5-2t)) < 0$; induction on j concludes.

Put $x^{(0)} = x$ and $y^{(0)} = y$, and suppose that $[T_j\mu(x^{(j)})e] = [\mu(y^{(j)})e]$ for some $j < n$. Lemma 4.9, applied to T_j with $q = q_j$ and $\kappa = M$, shows that $x^{(j)}$ and $y^{(j)}$ both begin with c_{j+1}. Writing $x^{(j)} = c_{j+1}x^{(j+1)}$ and $y^{(j)} = c_{j+1}y^{(j+1)}$ and multiplying by $\mu(c_{j+1})^{-1}$ gives $[T_{j+1}\mu(x^{(j+1)})e] = [\mu(y^{(j+1)})e]$. After n steps $x = p\,x^{(n)}$ and $y = p\,y^{(n)}$, and the relation for $T_n = K$ is the claim with $x_0 = x^{(n)}$ and $y_0 = y^{(n)}$. □

Theorem 4.11 (Decoding at every palindromic centre)**.** *For every palindrome p and all finite words x and y, the empty word included, the lines $[R_p\mu(x)e]$ and $[\mu(y)e]$ coincide if and only if $x \in \{ab\,p,\ ba\,p\}^*$ and y is obtained from x by exchanging every block. In that case $R_p\mu(x)e = \mu(y)e$ exactly.*

Proof. The implication from right to left and the exact equality are proposition 4.5: for a block word x and its exchange $\widehat{x}$, $K\mu(x) = \mu(\widehat{x})K$, and $Ke = Me$. For the converse we argue by induction on $|x| + |y|$. Suppose $[K\mu(x)e] = [\mu(y)e]$. If one of the words is empty, both are, by lemma 4.8, and the empty word is a block word equal to its exchange. Otherwise their first letters differ. If x begins with a, lemma 4.8 gives $x = ab\,x_1$, $y = ba\,y_1$ and $[K^{\mathsf{T}}\mu(x_1)e] = [\mu(y_1)e]$, and lemma 4.10 gives $x_1 = p\,x''$, $y_1 = p\,y''$ and $[K\mu(x'')e] = [\mu(y'')e]$. Since $|x''| + |y''| < |x| + |y|$, the induction hypothesis shows that x'' is a block word and y'' its exchange, so $x = ab\,p\,x''$ is a block word and $y = ba\,p\,y''$ is its exchange. If x begins with b, the relation is symmetric, because $K^2 = M^2 I$; the previous case applied to (y, x) gives $y = ab\,p\,y''$ and $x = ba\,p\,x''$ with $[K\mu(y'')e] = [\mu(x'')e]$, hence $[K\mu(x'')e] = [\mu(y'')e]$, and the induction hypothesis concludes in the same way. □

For a standard centre $p = \mathrm{Pal}(d)$ one has $ab\,p = \phi_d(ab)$ and $ba\,p = \phi_d(ba)$ by (3), so the theorem contains theorem 3.1 and shows moreover that no word outside the image of ϕ_d is related to a word column by R_d. The proof uses neither the standard letters U, V nor the cylinders of section 3; those cylinders are still needed in section 5, where they measure how far a column moves when the midpoint is displaced. The theorem has the following consequence for palindromes.

Corollary 4.12 (Classification at every centre)**.** *Let p, Z and Z' be palindromes. Then $M(Z) = M(Z')$ and $\rho(Z) + \rho(Z') = 2M(Z)\rho_p/M_p$ if and only if $Z = p\,X$ and $Z' = p\,\widehat{X}$ for a block word $X \in \{ab\,p,\ ba\,p\}^*$. In that case either $Z = Z' = p$, or Z is structured*

around p and Z' is its exchange, and the two words have the same numbers of letters a and b. In particular two distinct central words never have equal labels with root midpoint at the ratio of a palindrome.

Proof. Put $J_\tau = \left(\begin{smallmatrix} 1 & \tau \\ 0 & -1 \end{smallmatrix}\right)$ with $\tau = 2\rho_p/M_p$; by the proof of proposition 4.5, $P^{-1}R_pP = J_\tau$. The computation in the proof of lemma 4.1 shows that $J_\tau^{\mathsf{T}}\mathcal{G}(Z)J_\tau$ is symmetric of determinant one with first row $(M(Z), \tau M(Z) - \rho(Z))$, and a symmetric matrix of determinant one with positive upper-left entry is determined by its first row. So the two conditions on labels and roots hold if and only if $\mathcal{G}(Z') = J_\tau^{\mathsf{T}}\mathcal{G}(Z)J_\tau$, that is, since P is symmetric, $\mu(Z') = R_p^{\mathsf{T}}\mu(Z)R_p$. If this holds, then $\mu(Z')e = R_p^{\mathsf{T}}\mu(Z)e$ because $R_pe = e$, and lemma 4.10 gives $Z = p\,X$ and $Z' = p\,Y$ with $[K_p\mu(X)e] = [\mu(Y)e]$; by theorem 4.11, X is a block word and $Y = \widehat{X}$. Conversely, if $Z = p\,X$ and $Z' = p\,\widehat{X}$, then $K_p\mu(X) = \mu(\widehat{X})K_p$ and $K_p^2 = M_p^2 I$ give $\mu(\widehat{X}) = R_p\mu(X)R_p$, so $\mu(Z') = G\,R_p\mu(X)R_p = R_p^{\mathsf{T}}G\mu(X)R_p = R_p^{\mathsf{T}}\mu(Z)R_p$. Since Z is a palindrome and every block has length $|p| + 2$, comparing Z with its reversal shows that X has an even number $2k$ of blocks with $W_{2k+1-i} = \widehat{W_i}$; so either $k = 0$, or Z is structured around p and Z' is its exchange. Exchanging blocks does not change the letter counts. Finally a word structured with $k \geqslant 1$ has completed counts $(2k+1)\,c(p)$, which are not coprime, so it is not central. □

The centre in the corollary is determined by the pair: the fraction ρ_p/M_p is reduced, because $\rho_p^2 \equiv -1 \pmod{M_p}$, so it gives M_p and ρ_p, hence the Gram matrix of p and, the Cohn coding being injective, the palindrome p itself. The corollary contains theorem 4.2: when $Z = F_d(z_X)$ and $Z' = F_d(z_Y)$ and $p = \mathrm{Pal}(d)$, the block word X lies in $\phi_d(\{ab, ba\}^*)$, and the injectivity of F_d returns the pair $(\widetilde{w}w, \widetilde{\widehat{w}}\,\widehat{w})$. Remark 4.7 shows that the hypothesis on the midpoint cannot be dropped.

4.4. **Nested centres and shifted midpoints.** The quadruple of remark 4.7 raises two questions: why do four palindromes share a label, and why are the midpoints of two of its pairs shifted off every centre? The answer is that exchanges around nested centres compose. When the centre of an exchange is itself one of two exchanged palindromes, the words built around the two of them form a square of twins, and nesting r levels gives a cube with 2^r vertices. The shift of a midpoint is then the geometry of two reflections of the ratio axis, which compose to a translation.

It is convenient to name the words of section 4.2 by their blocks. A *block sequence* is a finite sequence $B = (B_1, \ldots, B_n)$ with $B_i \in \{ab, ba\}$; we write $\widehat{B} = (\widehat{B}_1, \ldots, \widehat{B}_n)$, and we call B *structured* when $n = 2k \geqslant 2$ and $B_{n+1-i} = \widehat{B}_i$ for every i. For a word q put

$$q\langle B\rangle = q\,B_1\,q\,B_2\,q\cdots q\,B_n\,q.$$

The words structured around a palindrome p are exactly the words $p\langle B\rangle$ with B structured, and the exchange of $p\langle B\rangle$ around p is $p\langle\widehat{B}\rangle$. For block sequences $X = (X_1, \ldots, X_n)$ and $Y = (Y_1, \ldots, Y_m)$ let

$$X * Y = (X,\, Y_1,\, X,\, Y_2,\, \ldots,\, X,\, Y_m,\, X)$$

be the block sequence in which the blocks of X are written $m + 1$ times, separated by $Y_1, \ldots, Y_m$; it has $(m+1)n + m$ blocks.

Lemma 4.13 (Reading through an inner centre)**.** *For every word q and all block sequences X and Y,*

$$(q\langle X\rangle)\langle Y\rangle = q\langle X * Y\rangle, \qquad \widehat{X * Y} = \widehat{X} * \widehat{Y}.$$

If X and Y are structured, then so is $X * Y$.

Proof. The word $Q = q\langle X\rangle$ begins and ends with q. In $Q\langle Y\rangle = Q\,Y_1\,Q\cdots Y_m\,Q$ every junction reads $\cdots X_n\,q\,Y_j\,q\,X_1\cdots$, so the whole word is q followed by the blocks $X_1,\dots,X_n,Y_1,X_1,\dots,X_n,Y_2,\dots,Y_m,X_1,\dots,X_n$, each of them followed by one copy of q. This is $q\langle X * Y\rangle$. The second identity holds block by block. For the last assertion let X have $n = 2k$ blocks and Y have $m = 2j$ blocks. Reading $X * Y$ backwards gives $(\overleftarrow{X}, Y_m, \overleftarrow{X},\dots,Y_1,\overleftarrow{X})$, where $\overleftarrow{X} = (X_n,\dots,X_1)$. Structure of X means $\overleftarrow{X} = \widehat{X}$, and structure of Y means $Y_{m+1-i} = \widehat{Y}_i$, so the reversed sequence is $(\widehat{X},\widehat{Y}_1,\widehat{X},\dots,\widehat{Y}_m,\widehat{X}) = \widehat{X} * \widehat{Y}$, which is $\widehat{X * Y}$ by the second identity. Its number of blocks, $(m+1)n+m$, is even and at least two. □

Theorem 4.14 (Cubes of nested exchanges)**.** *Let p be a palindrome, let $r \geqslant 1$, and let $B_1,\dots,B_r$ be structured block sequences, B_i with $2k_i$ blocks. For $s \in \{0,1\}^r$ put $B_i^s = B_i$ if $s_i = 0$ and $B_i^s = \widehat{B}_i$ if $s_i = 1$, and define*

$$Q_0^s = p, \qquad Q_i^s = Q_{i-1}^s\langle B_i^s\rangle \quad (1 \leqslant i \leqslant r), \qquad Z^s = Q_r^s.$$

For $1 \leqslant i \leqslant r$ let $f_i \in \{0,1\}^r$ have its entries equal to one exactly in the positions $i,\dots,r$, and write $s + f_i$ for the sum modulo two. Then:

(i) *the 2^r words Z^s are pairwise distinct palindromes, all with completed counts $(2k_1+1)\cdots(2k_r+1)\,c(p)$;*

(ii) *for every s and every i, the words Z^s and Z^{s+f_i} are exchanges of each other around the palindrome Q_{i-1}^s;*

(iii) *all 2^r words have the same Gram label M, any two of them are joined by at most r exchanges of the kind in* (ii)*, and the roots of the pair in* (ii) *satisfy $\rho(Z^s) + \rho(Z^{s+f_i}) = 2M\rho(Q_{i-1}^s)/M(Q_{i-1}^s)$.*

Proof. Every level is a centre. Define block sequences $R_r^s = B_r^s$ and $R_i^s = B_i^s * R_{i+1}^s$ for $1 \leqslant i < r$. We claim that $Z^s = Q_{i-1}^s\langle R_i^s\rangle$ for every i, and prove it by downward induction on i. For $i = r$ it is the definition $Z^s = Q_r^s = Q_{r-1}^s\langle B_r^s\rangle$. If $Z^s = Q_i^s\langle R_{i+1}^s\rangle$, then $Q_i^s = Q_{i-1}^s\langle B_i^s\rangle$ and lemma 4.13 give $Z^s = Q_{i-1}^s\langle B_i^s * R_{i+1}^s\rangle = Q_{i-1}^s\langle R_i^s\rangle$. By the last assertion of lemma 4.13 and induction, every R_i^s is structured. The same argument applied to the word Q_{i-1}^s in place of Z^s writes it as $p\langle R\rangle$ with R structured, or as p itself when $i = 1$; in both cases Q_{i-1}^s is a palindrome. Hence Z^s is structured around the palindrome Q_{i-1}^s.

The exchange is a flip of the last coordinates. The exchange of Z^s around Q_{i-1}^s is $Q_{i-1}^s\langle\widehat{R_i^s}\rangle$. Applying the second identity of lemma 4.13 repeatedly, $\widehat{R_i^s} = \widehat{B_i^s} * \widehat{R_{i+1}^s} = \cdots$ is the sequence $R_i^{s'}$ built from $s' = s + f_i$, the vector obtained by changing $s_i,\dots,s_r$. The first $i-1$ coordinates of s' are those of s, so $Q_{i-1}^{s'} = Q_{i-1}^s$, and the exchange is $Q_{i-1}^{s'}\langle R_i^{s'}\rangle = Z^{s'}$. This proves (ii).

Palindromes, distinctness and counts. The case $i = 1$ of the first step gives $Z^s = p\langle R_1^s\rangle$ with R_1^s structured, so Z^s is a palindrome. Let $s \neq s'$, let i be the first coordinate in which they differ, and put $Q = Q_{i-1}^s = Q_{i-1}^{s'}$. Every Q_j^s begins with Q_{j-1}^s, so Z^s begins with $Q\langle B_i^s\rangle$ and $Z^{s'}$ begins with $Q\langle\widehat{B_i^s}\rangle$; the two words differ in the letter that follows the first copy of Q, which is the first letter of a block in one and of its exchange in the other. By proposition 4.5, a word structured around a centre with $2k$ blocks has completed counts $2k+1$ times those of the centre; applied at the levels $1,\dots,r$ this gives the counts in (i).

Labels and roots. By proposition 4.5, two words exchanged around a palindrome have the same label, and their roots sum to twice the label times the ratio of the centre. The vectors $f_1,\dots,f_r$ are linearly independent modulo two, since the matrix with these rows is triangular with ones on the diagonal. Every vector of $\{0,1\}^r$ is therefore a sum of at

most r of them, and applying the corresponding exchanges one after the other joins Z^s to any other $Z^{s'}$. The label is therefore common to all 2^r words, which proves (iii). □

For $r = 2$ the cube is a square, and its two pairs that are not exchanges are the pairs of the quadruple whose midpoints are shifted.

Corollary 4.15 (The square and its shifted midpoints)**.** *Let p be a palindrome and W, V structured block sequences; put $P_1 = p\langle W\rangle$, $P_2 = p\langle\widehat{W}\rangle$ and*

$$Z_1 = P_1\langle V\rangle, \qquad Z_2 = P_1\langle\widehat{V}\rangle, \qquad Z_3 = P_2\langle V\rangle, \qquad Z_4 = P_2\langle\widehat{V}\rangle,$$

with common label M and roots $\rho_1, \dots, \rho_4$. The pairs (Z_1, Z_2) and (Z_3, Z_4) are exchanges around P_1 and P_2, and the pairs (Z_1, Z_4) and (Z_2, Z_3) are exchanges around p. The two remaining pairs satisfy $\rho_1 \neq \rho_2$ and

$$\frac{\rho_1 + \rho_3}{2M} = \frac{\rho_p}{M_p} + \frac{\rho_1 - \rho_2}{2M}, \qquad \frac{\rho_2 + \rho_4}{2M} = \frac{\rho_p}{M_p} - \frac{\rho_1 - \rho_2}{2M}.$$

If W and V have two blocks each, then $M(P_1) = M_p(9M_p^2+1)$, $M = M(P_1)(9M(P_1)^2+1)$ and $|\rho_1 - \rho_2| = 6M(P_1)$.

Proof. Take $r = 2$, $B_1 = W$ and $B_2 = V$ in theorem 4.14; then Z_1, Z_2, Z_3, Z_4 are $Z^{(0,0)}, Z^{(0,1)}, Z^{(1,0)}, Z^{(1,1)}$. The flip $f_2 = (0, 1)$ joins Z_1 to Z_2 around $Q_1^{(0,0)} = P_1$ and Z_3 to Z_4 around $Q_1^{(1,0)} = P_2$. The flip $f_1 = (1, 1)$ joins Z_1 to Z_4 and Z_2 to Z_3, both around $Q_0 = p$. By part (iii) of the theorem, $\rho_1 + \rho_4 = 2M\rho_p/M_p$ and $\rho_2 + \rho_3 = 2M\rho_p/M_p$. Hence

$$\rho_1 + \rho_3 = \rho_1 - \rho_2 + 2M\frac{\rho_p}{M_p}, \qquad \rho_2 + \rho_4 = \rho_2 - \rho_1 + 2M\frac{\rho_p}{M_p},$$

which are the displayed identities after division by $2M$. The words Z_1 and Z_2 are distinct palindromes with the same label, and a palindrome is determined by its label and root, because they determine the Gram matrix through its determinant and the Cohn coding is injective; so $\rho_1 \neq \rho_2$. If W and V have two blocks each, proposition 4.4 applied around p gives the label of P_1, and applied around P_1 it gives M and $\rho_1 - \rho_2 = \pm 6M(P_1)$. □

For $p = \varepsilon$ and $W = V = (ba, ab)$ one has $P_1 = baab$, $P_2 = abba$, $M(P_1) = 1130$, and the four words are those of remark 4.7, with $M = 1130 \cdot 11\,492\,101$. The two pairs that are not exchanges have their midpoints at $2/5 \pm 3390/M$: the shift is half the width $6 \cdot 1130$ of the rungs around *baab* and *abba*, divided by the label. Figure 2 draws this square above the rung of *abba* and *baab*.

When every block sequence has two blocks, the arithmetic of a cube is completely explicit. Put $M_i = M(Q_i^s)$, which by proposition 4.4 does not depend on s, and $N_i = 9M_i^2 + 1$ for $0 \leqslant i < r$, so that $M_0 = M_p$ and $M_i = M_{i-1}N_{i-1}$. The same proposition gives $\rho(Q_i^s) = N_{i-1}\rho(Q_{i-1}^s) \pm 3M_{i-1}$, with the sign + exactly when the first block of B_i^s is *ba*. Since $N_{i-1} \equiv 1$ modulo $M_{i-1} = M_pN_0 \cdots N_{i-2}$, the numbers $M_p, N_0, \dots, N_{r-1}$ are pairwise coprime with product M. Reducing the recursion, $\rho(Q_i^s) \equiv \rho(Q_{i-1}^s)$ modulo M_{i-1} and $\rho(Q_i^s) \equiv \pm 3M_{i-1}$ modulo N_{i-1}; since every later level keeps the residue modulo the label of the previous one,

$$\rho(Z^s) \equiv \rho_p \pmod{M_p}, \qquad \rho(Z^s) \equiv \pm 3M_{i-1} \pmod{N_{i-1}} \quad (1 \leqslant i \leqslant r),$$

and different s give different patterns of signs. In the correspondence between roots of -1 and Gaussian integers recalled before proposition 4.6, $N_{i-1} = (3M_{i-1})^2 + 1$ is the norm of $3M_{i-1} \pm i$, whose roots are $\pm 3M_{i-1}$. The 2^r twins of a cube therefore carry the Gaussian integers $\gamma_p \prod_{i=1}^{r}(3M_{i-1} \pm i)$, with γ_p the Gaussian integer attached to ρ_p, for all choices of signs, and the exchange around Q_{i-1}^s conjugates the factors of the levels $i, \dots, r$. This is the sign change of proposition 4.6, made explicit. Every coincidence of

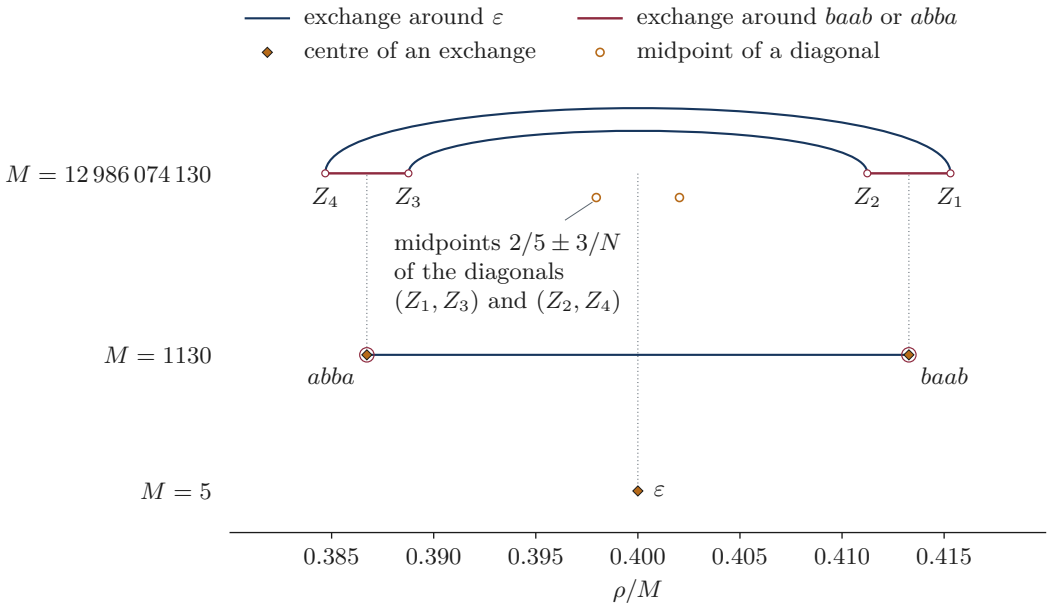


FIGURE 2. The square of corollary 4.15 for $p = \varepsilon$ and $W = V = (ba, ab)$, in the plane of the ratio ρ/M and the label, at the labels 5, 1130 and 12 986 074 130. The two lower levels are drawn to scale. At the top level Z_1 and Z_2 lie at $467/1130 \pm 3/N$ and Z_3 and Z_4 at $437/1130 \pm 3/N$, where $N = 9 \cdot 1130^2 + 1 = 11\,492\,101$; these offsets, about $2.6 \cdot 10^{-7}$, are drawn enlarged, all by the same factor. The short rungs are the exchanges around *baab* and *abba*, each centred above its centre; the arcs are the exchanges around the empty word, centred at $2/5$ like the rung below them. The diagonals (Z_1, Z_3) and (Z_2, Z_4) are not exchanges: their midpoints $2/5 \pm 3/N$ are the inner centre shifted by half the width of a short rung.

labels among palindromes below 10^{26} arises from such cubes, but not every coincidence does (remark 6.2), and section 6 takes up the question.

5. Displaced reflections and the size of a common ancestor

The ancestor-midpoint hypothesis has an exact algebraic meaning: it is the unique position where the reflection fixes the terminal column on both sides of the word equation. Moving the midpoint introduces a rank-one displacement. We first retain this displacement exactly, then combine the two first-letter cylinders with an integral rational-midpoint coordinate. The resulting bound applies to both parities of the common label. It is a necessary restriction on an equal-label pair, not a classification of all displaced pairs.

5.1. **The full two-sided equation.** Fix a directive d, put $w = \mathrm{Pal}(d)$, and retain the right standard map $F_d(z) = w\phi_d(z)$. Write

$$G = \mu(w), \quad \nu = \mu \circ \phi_d, \quad M_0 = e^{\mathsf{T}} G e, \quad \rho_0 = (Ge)_1, \quad r_0 = \rho_0/M_0.$$

The Gram matrix of the ancestor is PGP, with first row (M_0, ρ_0). Define two primitive integral columns

$$\omega = Ge = \begin{pmatrix} \rho_0 \\ M_0 - 2\rho_0 \end{pmatrix}, \qquad f = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \qquad a_0 = G^{-1} f = \begin{pmatrix} M_0 - 2\rho_0 \\ -\rho_0 \end{pmatrix}. \tag{48}$$

In particular $\omega^{\mathsf{T}} a_0 = 0$ and $f^{\mathsf{T}} a_0 = M_0$.

Proposition 5.1 (The moving reflection)**.** *Suppose the palindrome images $F_d(z_X), F_d(z_Y)$ have the same Gram label M and roots ρ_X, ρ_Y. Put*

$$\tau = \frac{\rho_X + \rho_Y}{M}, \quad \delta = \tau - 2r_0, \qquad R_0 = \frac{2ee^{\mathsf{T}}G - M_0 I}{M_0}.$$

Then, with $\nu_i = \nu(z_i)$,

$$\nu_Y = L_\tau \nu_X R_\tau, \tag{49}$$

$$R_\tau = R_0 + \delta e f^{\mathsf{T}}, \qquad L_\tau = R_0 + \delta a_0 \omega^{\mathsf{T}}. \tag{50}$$

The columns $v_i = \nu_i e$ satisfy the exact weighted-scale and displacement identities

$$\omega^{\mathsf{T}} v_X = \omega^{\mathsf{T}} v_Y = M, \qquad v_Y = R_0 v_X + \delta M a_0. \tag{51}$$

The right factor fixes e, whereas $L_\tau e = e + \delta M_0 a_0$.

Proof. The image Gram matrices have determinant one and the form

$$H_i = PG\nu_i P = \begin{pmatrix} M & \rho_i \\ \rho_i & (\rho_i^2 + 1)/M \end{pmatrix}.$$

For $J_\tau = \left(\begin{smallmatrix} 1 & \tau \\ 0 & -1 \end{smallmatrix}\right)$, direct multiplication gives $H_Y = J_\tau^{\mathsf{T}} H_X J_\tau$. Cancelling the invertible factors gives (49) with

$$R_\tau = P J_\tau P^{-1}, \qquad L_\tau = G^{-1} R_\tau^{\mathsf{T}} G.$$

At $\tau = 2r_0$ both are R_0; varying τ gives the two rank-one terms in (50). Since $R_\tau e = e$, the full equation gives $v_Y = L_\tau v_X$. The first Gram entry is $\omega^{\mathsf{T}} v_X = M$, proving (51). The terminal statement follows from $R_0 e = e$ and $\omega^{\mathsf{T}} e = M_0$. □

For a positive column v of slope $t = v_1/v_2$, use the reciprocal centered cusp coordinate

$$\eta(t) = \frac{t-2}{1 + r_0(t-2)} = \frac{M_0 f^{\mathsf{T}} v}{\omega^{\mathsf{T}} v}. \tag{52}$$

The denominator is positive because $0 < r_0 < 1/2$ and $t > 0$. The homogeneous expression also defines the coordinate whenever $\omega^{\mathsf{T}} v \neq 0$. The identities

$$\omega^{\mathsf{T}} L_\tau = \omega^{\mathsf{T}}, \qquad f^{\mathsf{T}} R_0 = -f^{\mathsf{T}}$$

give

$$\eta([L_\tau v]) = \xi - \eta([v]), \qquad \xi = \delta M_0^2. \tag{53}$$

Thus the terminal $\eta(e) = 0$ moves to ξ. For the two actual positive image columns,

$$\eta(t_X) + \eta(t_Y) = \xi.$$

This formula alone cannot exclude a pair of positive slopes: any such pair determines a value of ξ. The word equation must retain the common weighted scale in (51). Replacing L_τ by R_τ would erase precisely the information lost at a moving terminal.

5.2. **The actual common prefix and an integral midpoint direction.** Among palindromes in general, equal labels with a displaced midpoint do occur (remark 4.7), so centrality has to enter any restriction on displaced pairs. We now suppose $Z_X \neq Z_Y$ are central words with equal Gram label M. Let d be the longest common prefix of their unique directives. The following lemma ensures that this directive really supplies a symmetric matrix prefix.

Lemma 5.2 (The maximal common prefix)**.** *Let $Z_X \neq Z_Y$ be central words with equal Gram label M, let w_* be their maximal common symmetric prefix, so that $Z_i = w_* h_i \widetilde{w}_*$, and let d be the longest common prefix of their directives. Then both h_i have length at least two and begin with different letters, and*

$$w_* = \mathrm{Pal}(d) = w, \qquad Z_i = w h_i w = F_d(z_i), \tag{54}$$

where z_X, z_Y are nonempty central words beginning with different letters.

Proof. Every palindrome of length at least two has matrix at least A^2 entrywise: remove its two outer letters and use that every letter matrix is at least I. The matrix A^2 dominates I, A, B, strictly in at least one entry. Against the strictly positive column $\mu(w_*)^{\mathsf{T}} e$ its quadratic form therefore exceeds those of all length-zero and length-one cores, and the three short cores themselves have strictly ordered forms. Equal labels consequently cannot stop at a short core, and agreeing next letters would permit another symmetric stripping step; this proves the first assertion.

It follows that w_* is the full longest common word prefix. A directive extension $dx\cdots$ produces a central word beginning with $\mathrm{Pal}(d)x$, and neither directive can be a prefix of the other, since that would make one whole central word a prefix of the other, contrary to what has just been proved. This gives (54), in which z_i is the central preimage of the remaining directive and need not equal the literal core h_i. □

Write

$$PG = \begin{pmatrix} u & v \\ s & t \end{pmatrix}, \qquad ut - vs = -1, \quad u = 2s + t, \quad M_0 = 2u + v, \quad \rho_0 = u. \tag{55}$$

The four entries are nonnegative, with $u, v, s > 0$; the empty directive has $t = 0$. The symmetric factorization in (54) gives

$$H_i = (PG)\mu(h_i)(PG)^{\mathsf{T}}.$$

The two components of $\mu(h_i)(u, v)^{\mathsf{T}}$ are positive. Hence ρ_i/M is a strict positive weighted average of s/u and t/v, proving

$$\frac{t}{v} < \frac{\rho_i}{M} < \frac{s}{u}, \qquad \frac{s}{u} - \frac{t}{v} = \frac{1}{uv}. \tag{56}$$

The Gram label and root determine the complete Gram matrix, and the Cohn coding is injective. Thus distinct Z_i have distinct roots. Order them as $\rho_X < \rho_Y$, and set $\Sigma = \rho_X + \rho_Y$ and $\Delta\rho = \rho_Y - \rho_X > 0$. Both roots lie in $(0, M/2)$, so $0 < \Sigma < M$; none of these endpoints occurs.

Lemma 5.3 (A reduced midpoint without a parity assumption)**.** *Define*

$$c_0 = \gcd(\Sigma, 2M), \qquad Q = \frac{2M}{c_0}, \qquad \varepsilon_Q = \gcd(Q, 2), \qquad \kappa = \frac{\varepsilon_Q \Delta\rho}{Q}. \tag{57}$$

Then κ is a positive integer and there are unique coprime positive integers α, β satisfying

$$Q = \alpha u + \beta v, \qquad \frac{\Sigma}{c_0} = \alpha s + \beta t. \tag{58}$$

They obey

$$c_0 > \frac{2\kappa}{\varepsilon_Q} uv, \tag{59}$$

$$M_0\Sigma - 2\rho_0 M = c_0 n, \qquad n = \alpha - 2\beta. \tag{60}$$

For the distinct central words under consideration, $n \neq 0$.

Proof. The determinant-one Gram equations imply $\rho_X^2 \equiv \rho_Y^2 \equiv -1 \pmod{M}$, hence $M \mid \Sigma\Delta\rho$. Since $\gcd(\Sigma/c_0, Q) = 1$, this gives $Q \mid 2\Delta\rho$. Equivalently $Q/\varepsilon_Q \mid \Delta\rho$, proving positive integrality of κ without factoring M or assuming it odd.

The strict midpoint interval in (56) gives

$$\alpha = \frac{v\Sigma - 2Mt}{c_0} > 0, \qquad \beta = \frac{2Ms - u\Sigma}{c_0} > 0.$$

They are integers because c_0 divides both numerators. Inverting the unimodular frame proves (58), including uniqueness and coprimality. Moreover

$$\frac{\Delta\rho}{M} = \frac{2\kappa}{\varepsilon_Q c_0} < \frac{1}{uv},$$

which is (59). Substituting (58) into the left side of (60), and using (55), gives $c_0(\alpha - 2\beta)$.

Finally, $n = 0$ is precisely $\Sigma/(2M) = \rho_0/M_0$, the ancestor-midpoint case. Corollary 4.3 excludes that case for distinct central words. This last exclusion uses centrality; it is not asserted for arbitrary palindromic pairs. □

Put $r = \alpha/\beta$. Since $Q = \beta M_0\{1 + r_0(r-2)\}$, the integer-defect identity gives

$$\xi = \frac{2(r-2)}{1 + r_0(r-2)} = 2\eta(r), \qquad 2\eta(r) = \eta(t_X) + \eta(t_Y). \tag{61}$$

Thus the primitive direction is a nonlinear mean of the two actual column slopes, rather than a vector of letter counts. The root order also has a direct column interpretation. From $M = (u, v)v_i$, $\rho_i = (s, t)v_i$ and the frame identities, one obtains

$$\eta(t_i) = \frac{M_0^2 \rho_i}{M} - M_0\rho_0.$$

An a preimage has $t_i < 2$ and hence $\rho_i/M < r_0$; a b preimage has $t_i > 2$ and hence $\rho_i/M > r_0$. Therefore the a branch is precisely the smaller-root branch, consistently with the ordering above.

5.3. **Cancellation of the two parent cylinders.** Let $U = \nu(a)$ and $V = \nu(b)$, with $\operatorname{tr} U = 3p$, $\operatorname{tr} V = 3q$. We recall the notation of the decoder, but write its cusp centers as u_*, v_* to distinguish them from the prefix entries in (55). Put

$$a_* = p^{-2}, \quad b_* = q^{-2}, \quad L = u_* + v_* - 3 = \frac{M_0}{pq}, \quad \delta_* = (a_* - b_*)/L.$$

The standard-basis formulas in section 3 give

$$p \geqslant 1, \quad q \geqslant 2, \quad \frac{5}{2} \leqslant L < 3, \quad a_* + b_* = L(3 - L), \quad 2r_0 = u_* - v_* + \delta_*. \tag{62}$$

The preimages in (54) begin in opposite letters. Their cusp slopes $\zeta_i = 1/(t_i - 2)$ therefore lie in the two cylinders (29). Naming the a branch first, we may write

$$\begin{aligned} \zeta_X &= -u_* + e_U, & -\frac{5a_*}{12} &\leqslant e_U \leqslant \frac{12a_*}{25}, \\ \zeta_Y &= v_* + e_V, & -\frac{40b_*}{111} &\leqslant e_V \leqslant \frac{b_*}{2}. \end{aligned} \tag{63}$$

The centered coordinates $y_i = \zeta_i + r_0$ have the form

$$y_X = -\frac{L+3}{2} + \frac{\delta_*}{2} + e_U, \qquad y_Y = \frac{L+3}{2} + \frac{\delta_*}{2} + e_V.$$

The leading centers cancel in their reciprocal sum:

$$\xi = \frac{1}{y_X} + \frac{1}{y_Y} = \frac{\delta_* + e_U + e_V}{y_X y_Y}. \tag{64}$$

This cancellation explains why both parent labels enter the bound.

The ranges in (62)–(63) give

$$-\frac{a_*}{12}-\frac{422b_*}{555}\leqslant\delta_*+e_U+e_V\leqslant\frac{22a_*}{25}+\frac{b_*}{6}.$$

Every absolute coefficient here is at most $22/25$. The decoder's cusp orbit has $\zeta_X\leqslant-5/2$ and $\zeta_Y\geqslant 12/5$, while $0<r_0<1/2$. Hence $y_X<-2$, $y_Y>12/5$, and

$$|\xi|<\frac{11}{60}(a_*+b_*)\leqslant\frac{11}{48}.$$

Inverting (61) gives $r-2=\xi/(2-r_0\xi)$. Its denominator exceeds $181/96>0$, and therefore

$$\left|\frac{\alpha}{\beta}-2\right|<\frac{88}{905}(p^{-2}+q^{-2})<\frac{p^{-2}+q^{-2}}{10}\leqslant\frac{1}{8}. \tag{65}$$

No parity condition has entered either the cancellation or the rational direction.

Theorem 5.4 (A parent-sensitive bound in both parities)**.** *For two distinct central words with common Gram label M, let d be their maximal common directive and use the data above. Let*

$$\chi=\frac{p}{q}+\frac{q}{p},\qquad n=\alpha-2\beta\neq 0.$$

Then

$$\beta>\frac{10|n|p^2q^2}{p^2+q^2}=\frac{10|n|M_0}{\chi L},\qquad M_0^4<\frac{8\varepsilon_Q\chi(7+5\sqrt{2})}{25\kappa|n|}\,M. \tag{66}$$

In particular, bounded parent imbalance gives a fourth-power restriction on the common-ancestor label.

Proof. Since $|r-2|=|n|/\beta$, the first bound follows from (65). Set $\gamma=1+r_0(r-2)$. The same estimate gives $\gamma>15/16$, whence

$$Q=\beta\gamma M_0>\frac{10\gamma|n|}{\chi L}M_0^2>\frac{25|n|}{8\chi}M_0^2.$$

The prefix ratio u/v lies in $(\phi,1+\sqrt{2})$, where $\phi=(1+\sqrt{5})/2$: start at 2, and apply the two increasing Cohn maps, which preserve this interval. The function $x/(2x+1)^2$ decreases for $x>1/2$. Consequently

$$\frac{uv}{M_0^2}=\frac{u/v}{(2u/v+1)^2}>5\sqrt{2}-7.$$

Combining this with (59) and $M=c_0Q/2$ gives

$$M_0^4<\frac{\varepsilon_Q\chi L(7+5\sqrt{2})}{10\kappa\gamma|n|}\,M<\frac{8\varepsilon_Q\chi(7+5\sqrt{2})}{25\kappa|n|}\,M, \tag{67}$$

using $(5\sqrt{2}-7)^{-1}=7+5\sqrt{2}$. This also records the stronger data-dependent constant before its uniform simplification. □

The imbalance cannot be dropped. Along a constant a directive, one parent label stays equal to one, while $M_0=Lpq$ and $5/2\leqslant L<3$. Thus $\chi=q+1/q$ grows proportionally to M_0; the fourth-power bound then has the earlier cubic scale. The theorem distinguishes balanced parent scales from long one-sided directives without excluding either.

There is also a direct trace meaning for the integer defect. Put $H_0=(\rho_0^2+1)/M_0$, the lower-right entry of the ancestor Gram matrix. Since $(PGP)^{-1}H_i=P^{-1}\nu(z_i)P$, we have

$$\operatorname{tr}\nu(z_i)=\operatorname{tr}\big((PGP)^{-1}H_i\big)=H_0M+M_0\frac{\rho_i^2+1}{M}-2\rho_0\rho_i.$$

Completing a square gives

$$\operatorname{tr}\nu(z_i) = \frac{M}{M_0} + \frac{M_0}{M} + \frac{M_0}{M}\left(\rho_i - \frac{M\rho_0}{M_0}\right)^2.$$

Subtracting yields

$$\operatorname{tr}\nu(z_Y) - \operatorname{tr}\nu(z_X) = M_0\Delta\rho\,\delta = \frac{2\kappa n}{\varepsilon_Q}.$$

Thus the relative words do not themselves have equal traces when $n \neq 0$. Neither cancellation of the ancestor nor a projective moving-center argument supplies a new equal-label descent. The remaining problem is to combine literal centrality with the exact weighted-scale equality; the inequalities proved here are necessary conditions only.

6. Twins, ladders and what remains beyond the reflection axis

The classification has a simple mechanism. At the ancestor midpoint, the Gram reflection becomes a conjugacy that fixes the terminal column; the finite word is then recoverable from a rigid succession of paired blocks, and palindromicity turns the paired blocks into an odd common divisor of the completed counts. Proposition 4.5 shows that the constructive half of this mechanism works around every palindrome. This section asks whether the block exchange is the only source of equal labels among palindromes, records what an exact computation says about it, and isolates the step that a proof would need.

6.1. **A conjecture on twins.**

Call two distinct palindromes with the same Gram label *twins*. By proposition 2.6, twins are exactly the pairs of palindromes $av \neq av'$ without the factor bb for which the lengths $|\mathrm{Pal}(\theta(av))|$ and $|\mathrm{Pal}(\theta(av'))|$ coincide. Reutenauer and Vuillon proved that the Frobenius–Markov uniqueness conjecture is equivalent to the injectivity of this length on the central words $\mathrm{Pal}(av)$ [13, Theorem 1]. A search for coincidences among all palindromes without bb finds only words built around a palindromic centre, and it suggests the following conjecture.

Conjecture 6.1 (Structure of twins)**.** (i) *A palindrome has a twin if and only if it is structured around some palindrome.* (ii) *Any two twins are structured around a common palindrome.*

Part (ii) implies part (i), since a palindrome with a twin is then structured around the palindrome that it shares with its twin, and the *if* direction of part (i) is proposition 4.5. Neither part asks the twins of a palindrome to be its block exchanges, and they need not be: remark 4.7 shows twins that are not exchanges of each other, and remark 6.2 below shows twins that no chain of exchanges, each around its own centre, joins. Either part would imply the Frobenius–Markov uniqueness conjecture. Indeed, two distinct Markov occurrences with the same Markov number are twin central words, while a structured palindrome has completed counts divisible by $2k+1 \geqslant 3$ and is therefore not central.

An exact enumeration by computer of all 16 743 538 palindromes whose label does not exceed 10^{26} supports both parts of the conjecture. Exactly 1 719 of these labels are carried by more than one palindrome, 1 695 by two and 24 by four. Every palindrome involved is structured, every class of twins is connected by exchanges, and every exchange pair has its root midpoint at its centre. Every class of four is a square of corollary 4.15, so that every pair of twins whose midpoint is shifted off its centres is a diagonal of such a square, and the two palindromes of every class have the same numbers of letters a and b. Since the four words of a square are all structured around its inner centre (theorem 4.14), any two twins in this range are structured around a common palindrome. The enumeration is part of the supplementary material.

Remark 6.2 (Chains of exchanges do not suffice)**.** Above the range of the enumeration, twins need not be joined by exchanges. With $u = ab$ and $v = ba$, the structured block sequences

$$B = uvuuuvvuuuvuuvvuvvvuuvvvuv, \qquad B' = uvvuuuvuuuvvuvuuvvvuvvvuuv$$

of 26 blocks give two palindromes $Z = \varepsilon\langle B\rangle$ and $Z' = \varepsilon\langle B'\rangle$ of 52 letters with the same label

$$M(Z) = M(Z') = 19\,021\,048\,775\,294\,072\,282\,311\,853\,530\,130.$$

Testing every palindromic prefix as a centre shows that each of the four palindromes Z, Z', $\varepsilon\langle\widehat{B}\rangle$ and $\varepsilon\langle\widehat{B'}\rangle$ is structured around the empty word and around no other palindrome. The only exchange that applies to Z is therefore its exchange $\varepsilon\langle\widehat{B}\rangle$ around the empty word, and the only one that applies to that word leads back to Z; so the words that chains of exchanges reach from Z are Z and $\varepsilon\langle\widehat{B}\rangle$, and Z' is not among them. By corollary 4.12 the root midpoint of Z and Z' is then the ratio of no palindrome, so the pair is displaced, and it is not a diagonal of a square. Part (ii) of conjecture 6.1 holds for the four twins, since all of them are structured around the empty word, and all four have 26 letters a and 26 letters b. The supplementary material checks these statements in exact arithmetic.

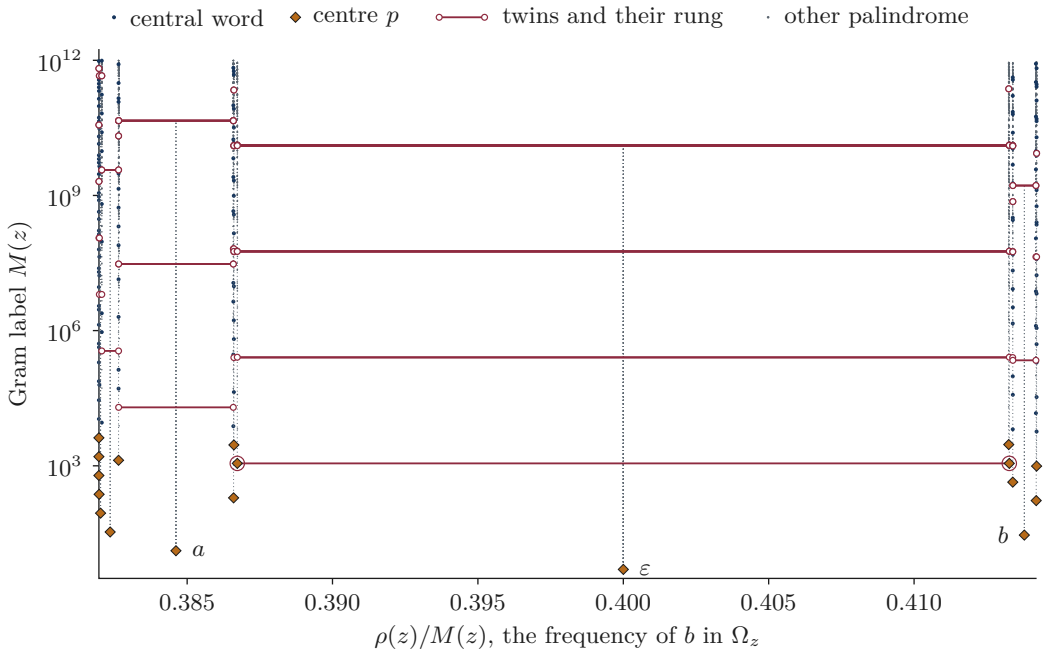


FIGURE 3. All palindromes with Gram label at most 10^{12}, placed by the ratio ρ/M and their Gram label M. The ratio $\rho/M = t/(2t+1)$ increases with the slope t of lemma 6.3, and it is the frequency of the letter b in Ω_z. Each horizontal segment joins two twins exchanged around a palindrome p (diamond), at the height of their common label; the dotted post shows that the segment is centred at the ratio ρ_p/M_p of p, which by lemma 6.3 is the lowest point between its ends. Ladders rise above the empty word, above a and above b. The two ringed diamonds, *abba* and *baab*, are twins themselves and centres of the exchanges at 12 986 074 130, the square of figure 2. No central word lies on a rung.

Figure 3 shows the twins in the plane of the ratio ρ/M and the label. They appear as the rungs of ladders, which are already visible in the alphabetical order of the directive words, and the following lemma explains why each rung stands on a palindrome of smaller label.

Lemma 6.3 (Ladders)**.** *For a word w let $t(w)$ be the slope of the column $\mu(w)e$. For every word q, the slopes of the words beginning with q lie in the interval $\mu(q)(\mathcal{J})$, which contains the slope of no other word. Consequently, if Z and Z' are twins and q is their longest common prefix, then every word whose slope lies strictly between $t(Z)$ and $t(Z')$ begins with q, the slope $t(q)$ is one of them, and q has strictly the smallest label among these words.*

Proof. By section 3, $A(\mathcal{J})$ and $B(\mathcal{J})$ lie in $\mathcal{J}$, on either side of 2, and both maps are increasing. A word qw' has slope $\mu(q)(t(w')) \in \mu(q)(\mathcal{J})$. A word that is not of this form either first differs from q at some letter, after a common prefix u, and then its slope lies in $\mu(u)\mu(c)(\mathcal{J})$ while $\mu(q)(\mathcal{J}) \subset \mu(u)\mu(c')(\mathcal{J})$ with $c \neq c'$, two disjoint intervals; or it is a proper prefix u of $q = uy$, and then its slope $\mu(u)(2)$ avoids $\mu(u)\mu(y)(\mathcal{J})$ because $2 \notin A(\mathcal{J}) \cup B(\mathcal{J})$. This proves the first assertion. Next, for a nonempty word $y = cy'$ the column $\mu(y)e - e$ has positive entries: by induction $\mu(y')e - e$ has nonnegative entries, and then $\mu(y)e - e = \mu(c)(\mu(y')e - e) + (\mu(c)e - e)$, where $\mu(c)$ has nonnegative entries and $\mu(c)e - e$ is $(3,2)^{\mathsf{T}}$ or $(10,4)^{\mathsf{T}}$. Since the row $e^{\mathsf{T}}\mu(q)$ has positive entries, $e^{\mathsf{T}}\mu(qy)e - e^{\mathsf{T}}\mu(q)e = e^{\mathsf{T}}\mu(q)(\mu(y)e - e) > 0$. In particular a twin cannot be a proper prefix of another twin. Hence $Z = q\,c\,Z_1$ and $Z' = q\,c'\,Z_1'$ with $c \neq c'$, say $c = a$. Both slopes lie in the interval $\mu(q)(\mathcal{J})$, and so does everything between them; and $t(Z) < \mu(q)(2) = t(q) < t(Z')$ because $t(aZ_1) < 2 < t(bZ_1')$. The words in between begin with q, and the same inequality of labels shows that q is the lowest among them. □

For an exchange pair around p the foot of the rung is p itself, and by proposition 4.5 the rung is centred on it in the ratio ρ/M, which is an increasing function of the slope t, so that the lemma applies to it unchanged. Since the lemma holds for every pair of twins, the ladders alone cannot decide the conjecture. In this picture, the conjecture implies that every palindrome with a twin has at least one rung centred on a palindrome, and section 6.2 shows that this consequence is equivalent to part (i).

6.2. **The midpoint statement.** Theorem 4.11 proves the decoder at every palindromic centre, and corollary 4.12 turns it into a classification: two palindromes with equal labels whose root midpoint is the ratio ρ_p/M_p of a palindrome p are exchanged around p. This makes part (i) of conjecture 6.1 equivalent to one statement about midpoints: *every palindrome with a twin has a twin whose root midpoint is the ratio of a palindrome.* Indeed, if a palindrome Z has a twin Z' whose root midpoint is ρ_p/M_p, then Z is structured around p by corollary 4.12; conversely a structured palindrome has its exchange as a twin with that midpoint, by proposition 4.5. No condition on a common prefix is needed: the corollary forces both palindromes to begin with p. Part (ii) asks in addition that the palindrome be common to the two twins, and in the same way it is equivalent to the statement that *any two twins Z and Z' have a common palindrome p such that each of them has a twin whose root midpoint with it is the ratio ρ_p/M_p.* This is the displaced problem of section 5, now posed for all palindromes. Remark 4.7 shows that the twin has to be chosen, since some twins are displaced, and the enumeration finds that every displaced pair is a diagonal of a square of corollary 4.15, that is, a composition of two exchanges; remark 6.2 shows that above its range a displaced pair need not be a composition of exchanges at all. For each pair the centre, when it exists, is unique, because the ratio of a palindrome determines it.

6.3. **Lengths of directing words.** In the Thue–Morse coding a coincidence of labels is a coincidence of the lengths $|\mathrm{Pal}(\theta(av))|$ and $|\mathrm{Pal}(\theta(av'))|$ of two central words, and it is natural to ask whether it forces the two directing words av and av' to have the same length. The question matters because the directing words of one length form a finite family, so that a positive answer would allow the coincidences to be studied one length at a time. This subsection gives the length its meaning in the Cohn coding, shows that twins joined by exchanges have directing words of the same length, and shows that for Markov words a positive answer is equivalent to the uniqueness conjecture.

Lemma 6.4 (The length of a directing word)**.** *For a word z put $M = e^{\mathsf{T}}\mu(z)e$ and $\rho = (\mu(z)e)_1$, which are the Gram label and the Gram root when z is a palindrome, and let $\sigma(z)$ be the sequence of integers obtained from z by replacing every letter a by $1,1$ and every letter b by $2,2$. Then $|F_a(z)| = |z|_a + 2|z|_b + 1$, and the regular continued fraction of M/ρ is*

$$\frac{M}{\rho} = [\,2;\ \sigma(z),\ 2\,].$$

In particular the sum of the partial quotients of M/ρ is $2|F_a(z)|+2$, the directive $\theta(F_a(z))$ has $2|F_a(z)|$ letters, and $|F_a(z)| = |azb|_b + |azb| - 2$.

Proof. The morphism ϕ_a sends the letter a to one letter and the letter b to two, which gives the first formula, and θ doubles every length. For a positive integer d put $D(d) = \left(\begin{smallmatrix} d & 1 \\ 1 & 0 \end{smallmatrix}\right)$, so that $P_1 = D(1)$ and $P = P_2 = D(2)$, with $A = P_1^2$ and $B = P_2^2$. Hence $P\mu(z)P = D(d_0)D(d_1)\cdots D(d_N)$, where $(d_0,\dots,d_N) = (2,\sigma(z),2)$. Let K denote the continuants, defined by $K() = 1$, $K(d_0) = d_0$ and $K(d_0,\dots,d_N) = d_0K(d_1,\dots,d_N) + K(d_2,\dots,d_N)$. The first column of $D(d_0)\cdots D(d_N)$ is $(K(d_0,\dots,d_N), K(d_1,\dots,d_N))^{\mathsf{T}}$: for a single factor it is $(d_N,1)^{\mathsf{T}} = (K(d_N),K())^{\mathsf{T}}$, and multiplying on the left by $D(d_0)$ replaces a first column $(x,y)^{\mathsf{T}}$ by $(d_0x+y,x)^{\mathsf{T}}$, which is the recurrence. On the other hand the first column of $P\mu(z)P$ is $P\mu(z)e$, because $Pe_1 = e$; with $\mu(z)e = (x,y)^{\mathsf{T}}$ it equals $(2x+y,x)^{\mathsf{T}} = (M,\rho)^{\mathsf{T}}$. Therefore $M = K(2,\sigma(z),2)$ and $\rho = K(\sigma(z),2)$. The quotient of these continuants is the continued fraction: $K(d_0)/K() = d_0$, and $[d_0;d_1,\dots,d_N] = d_0 + 1/[d_1;\dots,d_N] = d_0 + K(d_2,\dots,d_N)/K(d_1,\dots,d_N)$ by induction, which is $K(d_0,\dots,d_N)/K(d_1,\dots,d_N)$ by the recurrence. All partial quotients are positive and the last one equals two, so this is the regular continued fraction of M/ρ, which is unique. The pair $1,1$ adds 2 to the sum of the partial quotients and the pair $2,2$ adds 4, so the sum is $2 + 2|z|_a + 4|z|_b + 2 = 2|F_a(z)| + 2$. Finally $|azb|_b = |z|_b + 1$ and $|azb| = |z| + 2$. □

The length of the directing word is thus half the sum of the partial quotients of the label divided by the root, minus one: it counts, with multiplicity, the steps of the Euclidean algorithm on the pair (M,ρ). For a central word z the identity $M = K(2,\sigma(z),2)$ is the continued-fraction description of Markov numbers given by Frobenius [3, formula (6), p. 472]; the proof above does not use centrality. The value of a word in the matrices of Lagisquet, Pelantová, Tavenas and Vuillon [8] connects this length with their ordering of Markov numbers.

Proposition 6.5 (Lengths of directing words of Markov words)**.** *For a word w let $m(w) = (1,0)\,M^w\,(0,1)^{\mathsf{T}}$, where M^w is the product along w of $A_L = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ for the letter a and $B_L = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ for the letter b; this is the m-value of [8]. Then:*

(i) *$m(a\,F_a(z)\,b) = e^{\mathsf{T}}\mu(z)e$ for every word z;*
(ii) *if z is central, then $aF_a(z)b$ is the lower Christoffel word with $p = |z|_b + 1$ letters b and $q = |z| + 2$ letters a, the label $M(z)$ is the Markov number of the Farey fraction p/q, and $|F_a(z)| = p + q - 2$;*

(iii) *the Frobenius–Markov uniqueness conjecture holds if and only if any two central words with the same label have directing words of the same length; the same holds when the length of the directing word is replaced by the number of letters b of the word, or by its length.*

Proof. Part (i). Put $X = \left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$. Direct multiplication gives

$$XAX^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = A_L, \qquad XBX^{-1} = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} = B_L A_L.$$

Since $\phi_a(a) = a$ and $\phi_a(b) = ba$, this says $M^{\phi_a(c)} = X\mu(c)X^{-1}$ for both letters c, and both sides are multiplicative in the word, so $M^{\phi_a(w)} = X\mu(w)X^{-1}$ for every word w. As $aF_a(z)b = a\,a\,\phi_a(z)\,b$,

$$m(aF_a(z)b) = (1,0)\,A_L^2 X\,\mu(z)\,X^{-1} B_L \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Now $(1,0)A_L^2 = (2,3)$ and $(2,3)X = (2,1) = e^{\mathsf{T}}$, while $B_L(0,1)^{\mathsf{T}} = (1,1)^{\mathsf{T}}$ and $X^{-1}(1,1)^{\mathsf{T}} = (2,1)^{\mathsf{T}} = e$. This proves (i).

Part (ii). If $z = \mathrm{Pal}(v)$, then (2) gives $F_a(z) = \mathrm{Pal}(av)$, a central word, so $aF_a(z)b$ is a lower Christoffel word [2, Section 3]. The morphism ϕ_a keeps the number of letters b and adds one letter a for each letter of z; hence the word $aF_a(z)b$ has $|z|_b + 1 = p$ letters b and $2 + |z|_a + |z|_b = |z| + 2 = q$ letters a, and $p < q$. A lower Christoffel word is determined by its letter counts, which are coprime, so $aF_a(z)b$ is the word written $c(p,q)$ in [8], and by part (i) and [8, Proposition 3] its value $M(z)$ is the Markov number of the Farey fraction p/q; we write $m_{p/q}$ for that Markov number. By lemma 6.4, $|F_a(z)| = p + q - 2$. Conversely, every reduced fraction $0 < p/q < 1$ arises from exactly one central z. Write $c(p,q) = a\,w\,b$ with w central; w is not empty, since it has $q - 1 \geqslant 1$ letters a. If the directive of w began with b, then $w = F_b(w') = b\,\phi_b(w')$ with $\phi_b(a) = ab$, $\phi_b(b) = b$: every letter a of w would be followed by a letter b, and w would have more letters b than letters a. Since w has $q - 1 > p - 1$ letters a, its directive begins with a, so $w = F_a(z)$ for a central z, unique because F_a is injective.

Part (iii). The uniqueness conjecture is equivalent to the injectivity of $p/q \mapsto m_{p/q}$ on the reduced fractions of $(0,1)$, since the Markov tree indexes by these fractions the normalized Markov triples with distinct entries; by part (ii) this is the injectivity of the label on central words, and it makes the three statements vacuously true. Conversely suppose that two distinct central words have the same label, and let $p/q \neq p'/q'$ be their fractions. Write $m_{n,d}$ for the least value of m on the lattice paths from $(0,0)$ to (d,n); by [8, Proposition 21] it is attained at $c(n,d)$, so $m_{p,q}$ and $m_{p',q'}$ are the two equal labels. By [8, Theorem 2], for integers $j < k$,

$$m_{j,k} < m_{j,k+1}, \qquad m_{j,k} < m_{j+1,k}, \qquad m_{j+1,k} < m_{j,k+1}.$$

If $p+q = p'+q' = s$, then $p \neq p'$, say $p' < p$; for $p' \leqslant j \leqslant p-1$ one has $j < q-1 < s-j-1$, so the third inequality applied to $j < s - j - 1$ gives $m_{j+1,s-j-1} < m_{j,s-j}$, and chaining from $j = p - 1$ down to $j = p'$ gives $m_{p,q} < m_{p',q'}$, a contradiction. If $p = p'$, then $q \neq q'$, say $q < q'$, and the first inequality applied to $p < k$ for $q \leqslant k < q'$ gives $m_{p,q} < m_{p,q'}$. If $q = q'$, then $p \neq p'$, say $p < p'$, and the second inequality applied to $j < q$ for $p \leqslant j < p'$ gives $m_{p,q} < m_{p',q}$. Each case contradicts the equality of the labels, so a coincidence of labels among distinct central words changes the length of the directing word, the number of letters b, and the length. Each of the three statements therefore implies the uniqueness conjecture. □

Nothing in the proposition is deep, and most of it is known. Part (i) is, in our normalization, the identity $g(w) = m(aG(w)ab)$ of [8, Appendix B], where $G(a) = a$ and

$G(b) = ab$, so that $aG(w)ab = aF_a(w)b$, and where g is the function of Aigner's book, which is the label $e^{\mathsf{T}}\mu(w)e$ written in another basis. Part (iii) is an immediate consequence of the ordering theorems, and Lee, Li, Rabideau and Schiffler prove more: $m_{p/q} = m_{p'/q'}$ with $(p, q) \neq (p', q')$ forces $-5/4 < (p - p')/(q - q') < -8/7$ [10, Corollary 1.5]. Since $|F_a(z)| = p + q - 2$, a coincidence would change the length of the directing word by more than a seventh and less than a quarter of the change of q, and never by zero. The point of the proposition is the identification it makes: the directing word of the Thue–Morse coding is the central factor of the Christoffel word of the fixed-sum theorem, so that the length question is an equivalent form of the uniqueness conjecture.

Three consequences place the proposition. First, twins joined by exchanges have the same numbers of letters a and b, because an exchange does not change them, and so do the two twins of remark 6.2, which no exchange joins; if all twins had the same letter counts, they would have directing words of the same length, and by the proposition this alone would already imply uniqueness. Second, the equality of letter counts is a theorem on every reflection axis: by corollary 4.12, two twins whose root midpoint is the ratio of a palindrome are exchanged around it, so they have the same numbers of letters a and b and directing words of the same length; in the enumeration up to 10^{26} every class of twins has a single pair of letter counts. Third, in the language of [8], part (ii) says that their Christoffel word $c(p, q)$ is $aF_a(z)b$: the directing word of the Thue–Morse coding is its central factor, two letters shorter. Their fixed-sum theorem says that two distinct lower Christoffel words with more letters a than letters b and of the same length carry different Markov numbers; the length question asks for the complementary statement, that a coincidence of Markov numbers keeps the length, and the two statements together are the uniqueness conjecture.

6.4. **Off the axis.** Moving the midpoint breaks the reflection in an explicit way. The two matrix factors of proposition 5.1 remain involutions, but they differ, and the left factor moves the terminal column by a rank-one displacement. The integer defect and the parent-sensitive bound of theorem 5.4 give coordinates for this remaining problem: they show that a balanced common ancestor of two distinct central twins must be small compared with their label. They do not bound the two directive tails or prevent early branching. A proof of uniqueness along these lines would have to show that central words cannot realize a displaced midpoint, and centrality must enter there, since displaced twins exist among palindromes. The displaced twins found by the enumeration are all diagonals of the squares of corollary 4.15: each pair is a composition of two exchanges, and its root is translated by twice the label times the difference of the two centre ratios. Above that range displaced twins of another kind appear (remark 6.2); conjecture 6.1 asks of them only that they share a centre.

Supplementary material

The supplementary material accompanying this article contains the exact computations used in the proofs of theorems 3.1, 4.2 and 5.4, the worked examples of table 2, the identities and constants of the decoder at every palindromic centre in section 4.3, the nested cubes of section 4.4, the enumeration of twins of section 6, the two twins of remark 6.2, the identities of section 6.3, the data of the three figures, and instructions for repeating the verification.

Code and data availability

The source code and the data supporting this article, with the instructions that repeat every calculation, are contained in the supplementary material of this article, which is available, together with that of the other articles of this series:

https://github.com/dutykh/fibonacci-pell-nearest-gaps/

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During the preparation of this work the authors used Codex (OpenAI) and Claude Code (Anthropic) to assist with the research, the computations and the writing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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